Pre-Class Course Questionnaire

Name (optional):

- What is your current level of familiarity with the concept of "natural infrastructure"?

  Limited Knowledge [ ] Some Knowledge [ ] Moderate Knowledge [ ]
  Advanced Knowledge [ ] Expert Knowledge [ ]

- Have you had any previous experience or training in natural infrastructure or a related field?

  None [ ] Limited [ ] Some [ ] Moderate [ ] Extensive [ ]

Before participating in the field class, please rate your expectations on the following statements using a scale from Strongly Disagree to Strongly Agree:

- I expect the field class to contribute to my understanding of natural infrastructure and its role in water resource management.

  Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree [ ]

- I expect to achieve the goals and learning objectives I have set for myself before the field class.

  Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree [ ]

- I expect the fieldwork activities to enhance my practical skills in assessing and designing natural infrastructure.

  Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree [ ]

- I anticipate that collaborating with professionals, experts, and stakeholders during the field class will contribute to my understanding of complex issues related to natural infrastructure.

  Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree [ ]

- I expect to experience personal and professional growth during this field class, and I believe my perspective on the value of natural infrastructure will evolve.

  Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree [ ]

- I expect the field class to provide a safe and engaging learning environment.

  Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree [ ]

- I anticipate that the field class will offer opportunities for networking and building professional connections.

  Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree [ ]

Post-Class Course Questionnaire

Name (optional):

Please describe some significant observations during the field class that stood out to you.

1.

2.

Were there any unexpected discoveries during the field class that influenced your thinking?

1.

2.

On a scale from Strongly Disagree to Strongly Agree, please tick one of the following for each statement:

- The field class contributed to my understanding of natural infrastructure and its role in water resource management.

  Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree [ ]

- I achieved the goals and learning objectives I set before the field class.

  Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree [ ]

- The fieldwork activities enhanced my practical skills in assessing and designing natural infrastructure.

  Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree [ ]

- Collaborating with professionals, experts and stakeholders during the field class contributed to my understanding of complex issues related to natural infrastructure.

  Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree [ ]

- I experienced personal and professional growth during this field class and my perspective on the value of natural infrastructure evolved.

  Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree [ ]

- The field class fostered a safe and engaging learning environment.

  Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree [ ]

- The field class provided adequate opportunities for networking and building professional connections

  Strongly Disagree [ ] Disagree [ ] Neutral [ ] Agree [ ] Strongly Agree [ ]

Do you have any suggestions or recommendations for improving future iterations of this field class?

1.

2.

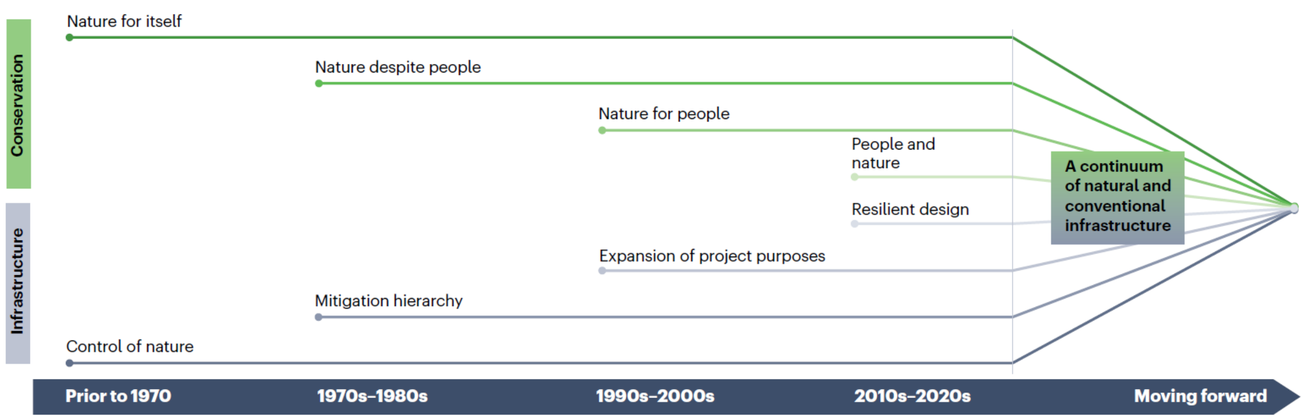
Day 1: Handout

What do you KNOW about Natural Infrastructure?	
What do you WANT to know about Natural Infrastructure?	
What did you LEARN about Natural Infrastructure today? (end of the day)	

Themes for today:

- Overview of natural infrastructure (NI) and nature-based solutions (NBS)
- Gathering the basic terminology from physical, ecological, and social sciences.
- Developing the historical context of a system: social, physical, and ecological legacies of the past; deeper view of time; reversibility; shifting baselines.

Converging arcs of biodiversity conservation and infrastructure (McKay et al. 2023)



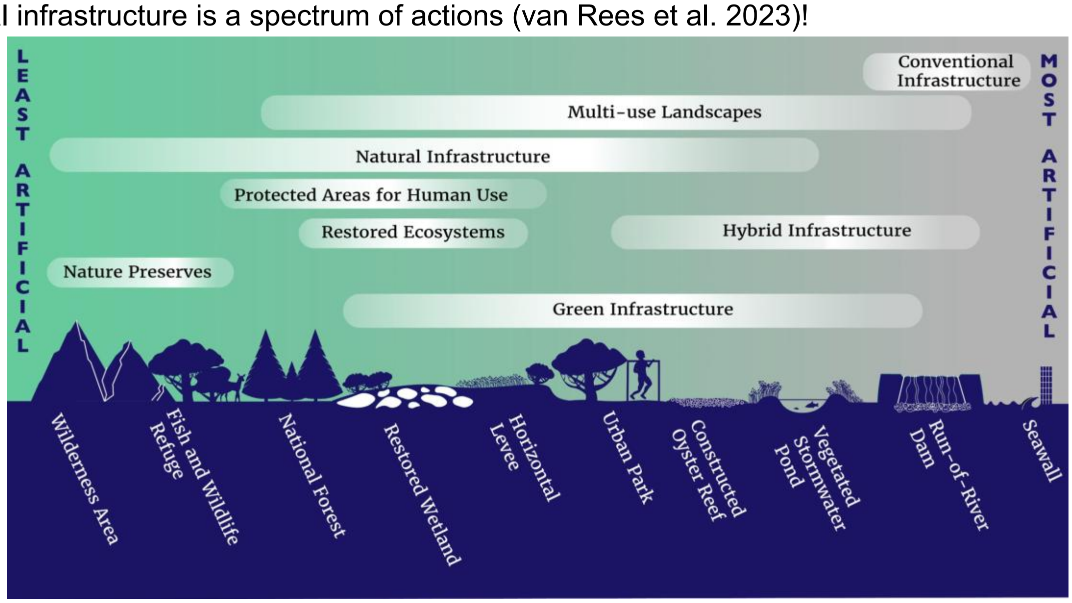
A paradigm shift is underway in the approach to planning infrastructure:

A Conventional Framing of Infrastructure	An Emergent Approach to Infrastructure
Infrastructure performs engineering functions and services	Infrastructure performs engineering functions and services
Fewer objectives centered on an organization's mission	Many objectives with a multi-purpose framing
Independent funding => independent projects	Leveraged funding => systems outcomes
Ecosystems treated as a constraint	Ecosystems included in the goals
Social benefits are happy coincidences	Social benefits are part of the design
Technical experts from a few disciplines develop alternatives	Technical experts lead alternative development, but other voices participate
Typically leads to more conventional, "grey" infrastructure	**Merges conventional infrastructure with natural systems**

Defining natural infrastructure

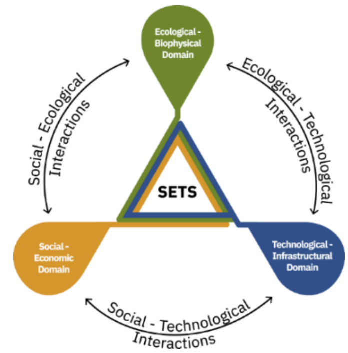
- UNEP (2022): Nature-Based Solution (NbS) are "actions to protect, conserve, restore, sustainably use and manage natural or modified terrestrial, freshwater, coastal and marine ecosystems, which address social, economic and environmental challenges effectively and adaptively, while simultaneously providing human well-being, ecosystem services and resilience and biodiversity benefits"
- Infrastructure Investment and Jobs Act (2021): Natural infrastructure "…uses, restores, or emulates natural ecological processes…is created through the action of natural, physical, geological, biological, and chemical processes over time; is created by human design, engineering, and construction to emulate or act in concert with natural processes; or involves the use of plants, soils, and other natural features..."
- In Practice, NI is (loosely) defined by several common characteristics, not all of which may be met by any particular feature: (1) Performs infrastructure services, (2) Consists, at least in part, of natural or living materials, (3) Intentionally provides environmental and social benefits beyond typical purposes, and (4) Enhances resilience through self-adjustment.

Natural infrastructure is a spectrum of actions (van Rees et al. 2023)!



Socio-Ecological-Technical Systems (SETS figure, McPhearson et al. 2021, *npj Urban Sustainability*)

- Any view of nature-based solutions inevitably intersects social, ecological, and technical systems
- These systems have many parts, but the interactions are where the real complexity lies.

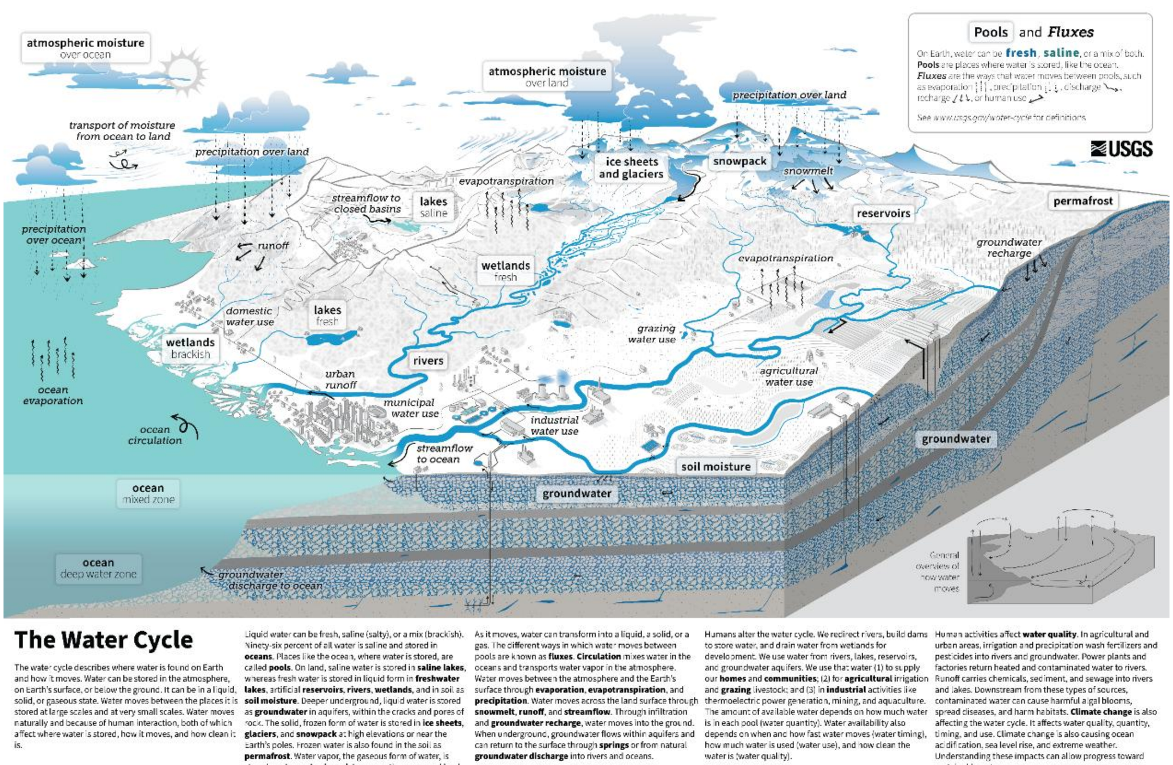


Principles of using a systems approach (de Vries et al. 2021, Chapter 4 of the International Guidelines for NNBF):

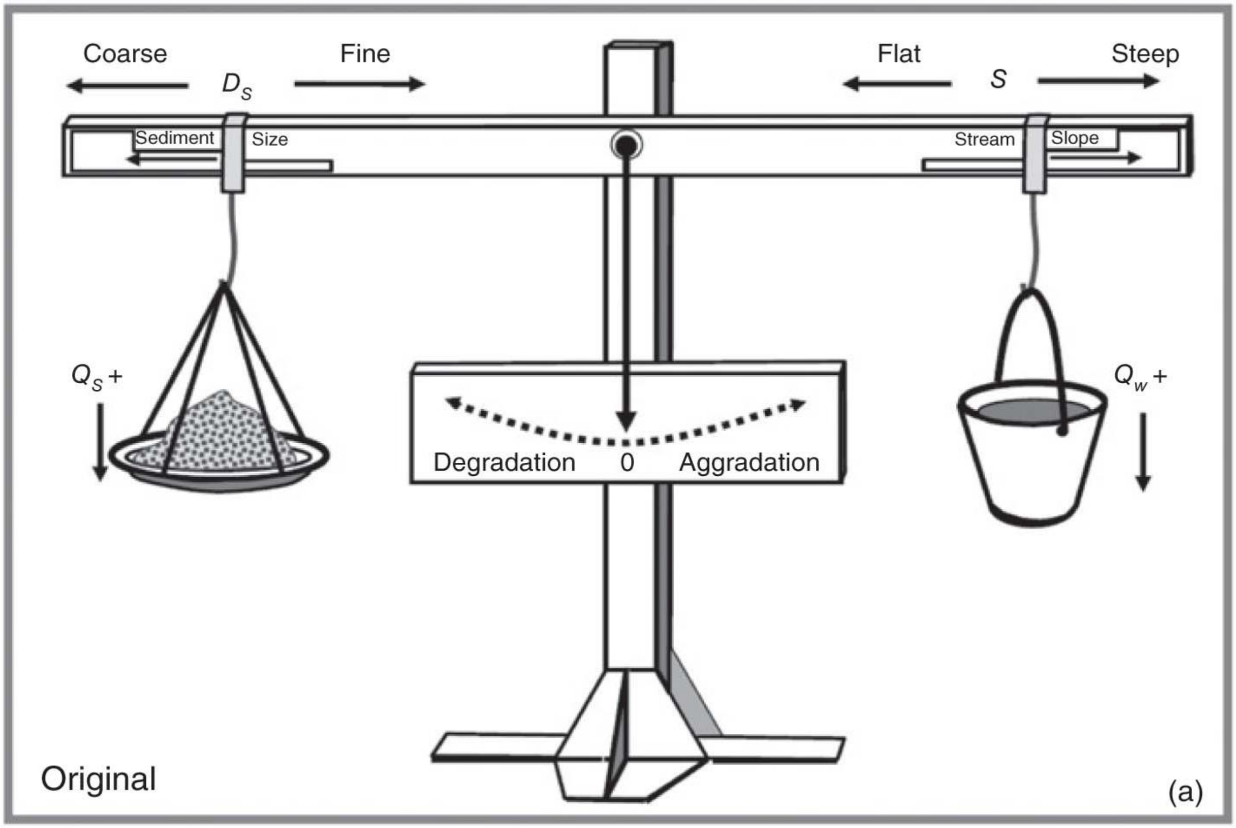
- Life safety is where flood risk management begins, but not where it ends.
- Flood risk management is nested in an interconnected socio-ecological landscape.
- Sustainable systems are resilient to disturbances.
- System-wide planning is a process worth the investment.
- Short-term benefits are balanced with long-term outcomes.

Physical systems and processes:

- Hydrology – water moving to the channel
- Hydraulics – water moving in the channel
- Fluvial geomorphology – how water shapes the earth (and how geology drives H&H)
- Biogeochemistry – the multitude of processes that affect the fate and transport of chemical constituents (and ultimately "water quality")
- Engineering (not covered here) – uses H&H, geomorphology, and biogeochemistry to move dirt or water for a purpose.

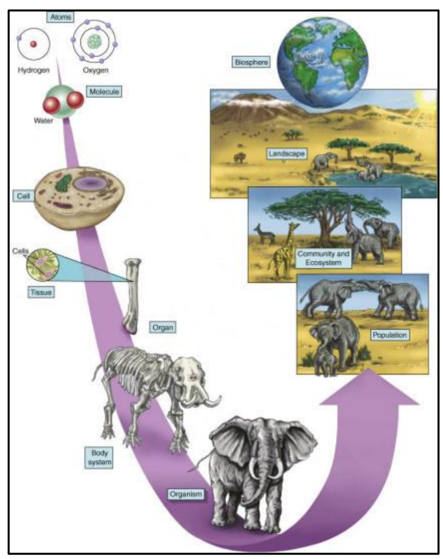


Lane's Balance is a conceptual model of qualitative channel response based on Sediment load (Qs), Sediment size (Ds), Discharge (Q), and Channel Gradient (S), where Qs Ds ~ Q S.



Ecosystems:

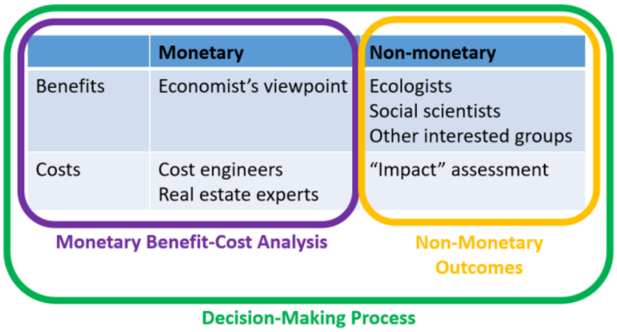
- Ecological systems are structured hierarchically.
- Population: a group of organisms of the same species occupying a particular space
- Community: a group of populations interacting in a particular space
- Ecosystem: a biotic community and its abiotic environment functioning as a system



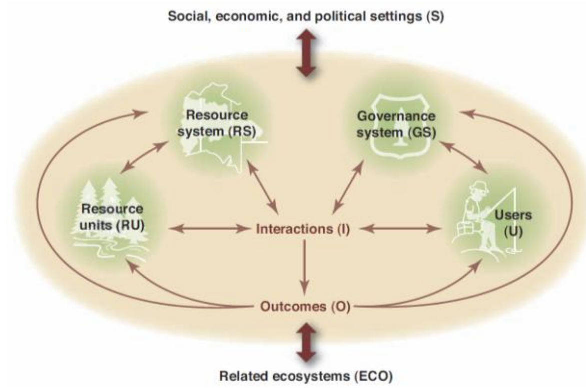
Jargon of ecological outcomes:

- Ecosystem structure: "refers to both the composition of the ecosystem (i.e., its various **parts**) and the physical and biological organization defining how those parts are organized" (Heal et al. 2005).
- Ecosystem function: "describes a process that takes place in an ecosystem as a result of the **interactions** of plants, animals, and other organisms in the ecosystem with each other or their environment" (Heal et al. 2005).
- Ecosystem service: "the benefits **people** obtain from ecosystems" (Millennium Ecosystem Assessment 2005).

A simple structure for benefits quantification



A nuanced model of common pool resource management (Ostrom 2009).



IAP2 Spectrum of Public Participation

iap2 international association for public participation

IAP2's Spectrum of Public Participation was designed to assist with the selection of the level of participation that defines the public's role in any public participation process. The Spectrum is used internationally, and it is found in public participation plans around the world.

INCREASING IMPACT ON THE DECISION

	INFORM	CONSULT	INVOLVE	COLLABORATE	EMPOWER
PUBLIC PARTICIPATION GOAL	To provide the public with balanced and objective information to assist them in understanding the problem, alternatives, opportunities and/or solutions.	To obtain public feedback on analysis, alternatives and/or decisions.	To work directly with the public throughout the process to ensure that public concerns and aspirations are consistently understood and considered.	To partner with the public in each aspect of the decision including the development of alternatives and the identification of the preferred solution.	To place final decision making in the hands of the public.
PROMISE TO THE PUBLIC	We will keep you informed.	We will keep you informed, listen to and acknowledge concerns and aspirations, and provide feedback on how public input influenced the decision.	We will work with you to ensure that your concerns and aspirations are directly reflected in the alternatives developed and provide feedback on how public input influenced the decision.	We will look to you for advice and innovation in formulating solutions and incorporate your advice and recommendations into the decisions to the maximum extent possible.	We will implement what you decide.

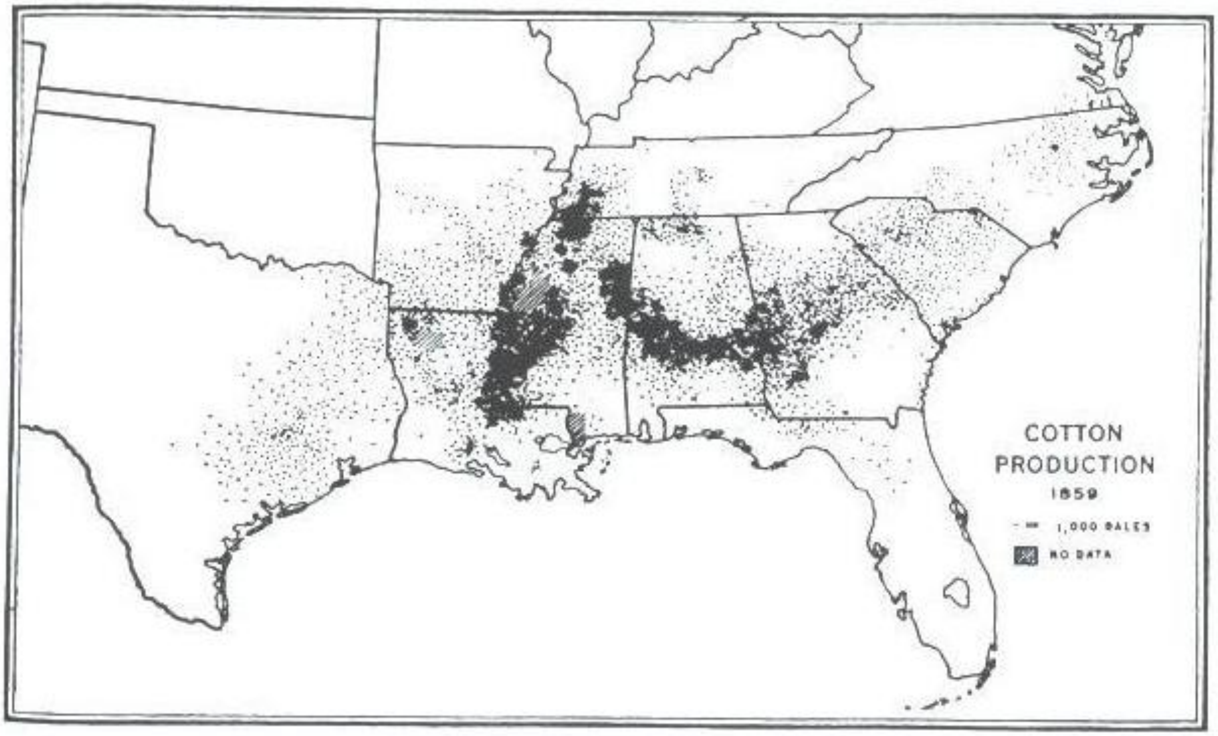


Figure: Gray (1933), Richter and Markewitz (2001)

Picture shared by Rhett Jackson

Day 1: Exercise

White Dam Conceptual Model

In teams of 3-4 people, develop a conceptual model of the White Dam system. The model can take any format (i.e., sketch, box-and-arrow, diagram), but it should seek to convey your understanding of the system and "tell the story" of the project. It may be constructive to use the steps for conceptual model development from Fischenich (2008). Discuss with your team how management objectives are featured in the model.

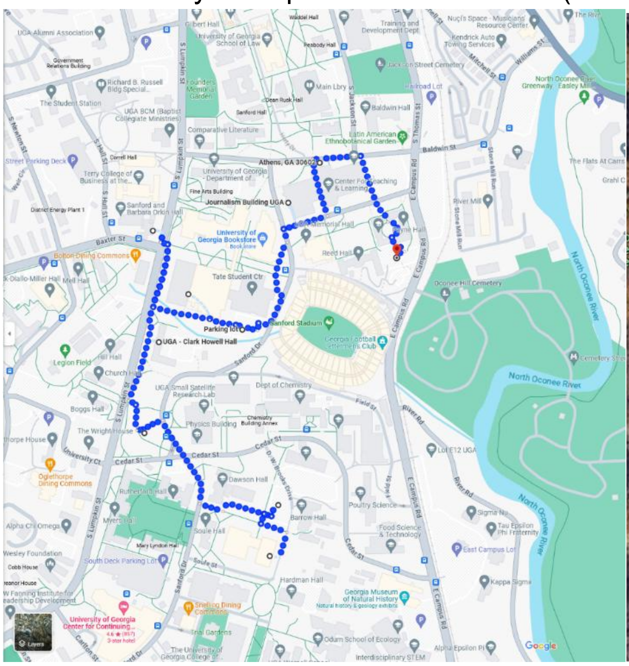
Steps for Conceptual Model Development (from Fischenich 2008)	**How you addressed them relative to White Dam**
1) State the model objectives.	
2) Bound the system of interest.	
3) Identify critical model components within the system of interest.	
4) Articulate the relationships among the components of interest.	
5) Represent the conceptual model.	
6) Describe an expected pattern of model behavior.	
7) Test, review, and revise as needed.	

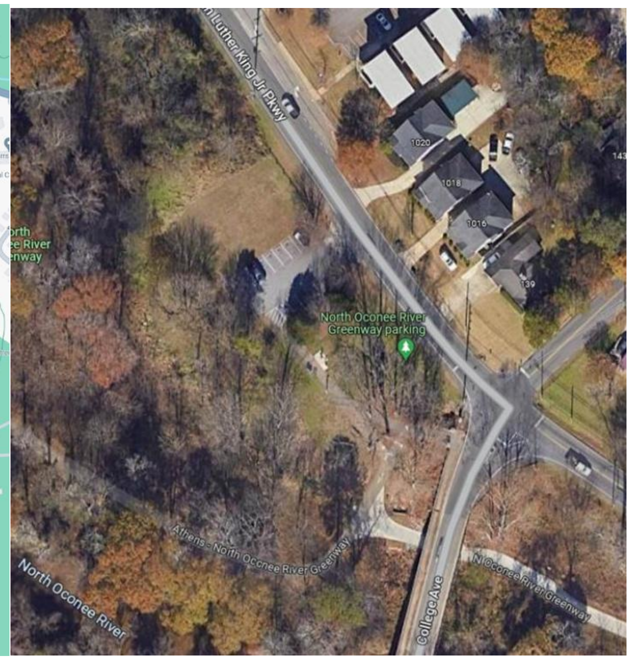
Day 2: Morning Handout

Themes for today:

- Urban / suburban land use effects on watershed processes
- Urban green infrastructure / natural infrastructure features and strategies
- Management of urban streams, floodplain corridors and riparian zones
- Equity considerations in urban NI
- Social benefits of urban NI

Plan for the day: Campus stormwater walk (morning, left). Greenway planning (afternoon, right).





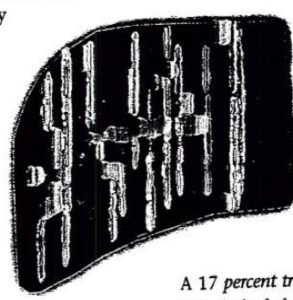
Texas City Relies on Tree Canopy to Reduce Runoff

Trees may be saving the city of Garland, Texas, more than $5 million a year in its measures to manage storm water, improve air quality, and save energy. American Forests, a nonprofit conservation organization in Washington, D.C., analyzed the trees' economic benefits in a study released earlier this year.

The city of Garland asked American Forests to conduct the study to address U.S. Environmental Protection Agency (EPA) water quality requirements. The city had been using innovative approaches to storm-water management for at least a decade. To satisfy the EPA, the city needed to demonstrate that the policies it had in place could reduce storm-water discharge enough to make an expensive structural solution unnecessary.

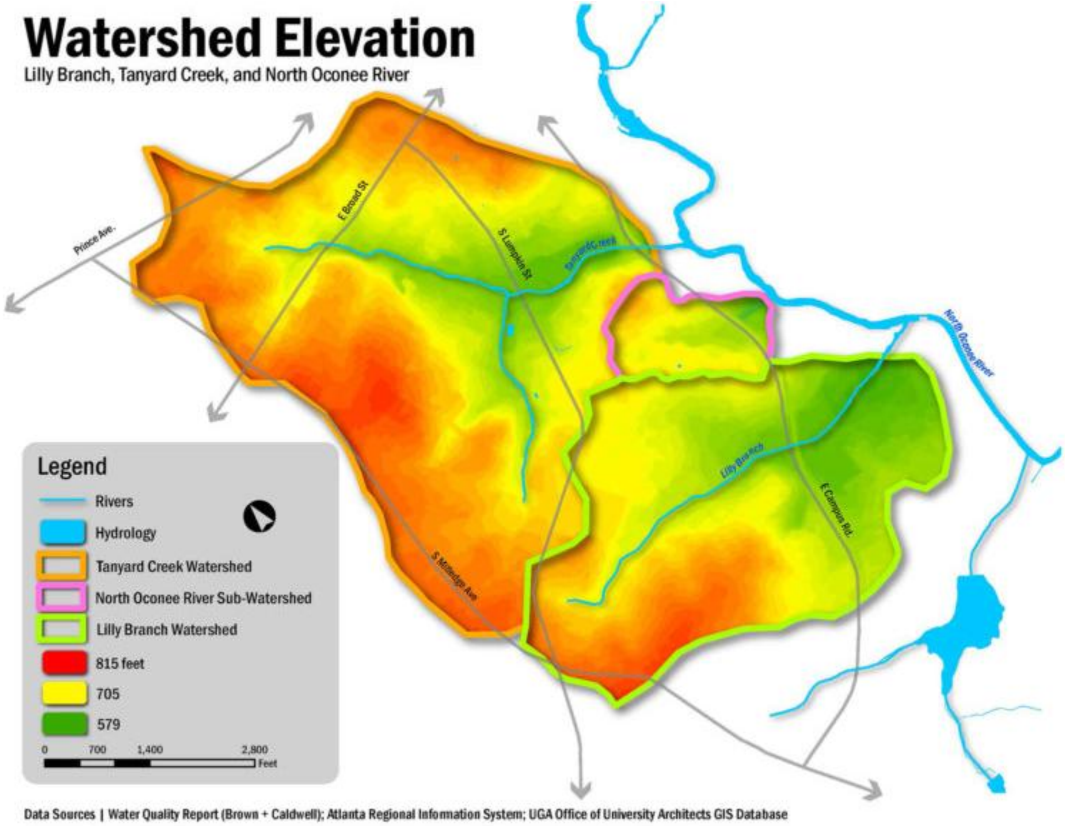
American Forests calculated that Garland's tree canopy reduces storm-water runoff by 19 million cu ft (540,000 m^3) during a major storm. The amount of runoff was multiplied by the cost of building storm-water containment facilities, estimated at $2/cu ft ($71/m^3).

Although Garland has a tree cover of only 11 percent—less than average for cities of its size—American Forests estimated that the trees save Garland $2.8 million annually in infrastructure costs. In addition, the study showed that because trees reduce air pollution and provide shade, they save the city $2.5 million annually in air quality costs and residential energy bills. ▼



A 17 percent tree canopy at this 8 acre (3 ha) site helped reduce runoff by nearly 7 percent.

UGA Campus Watersheds



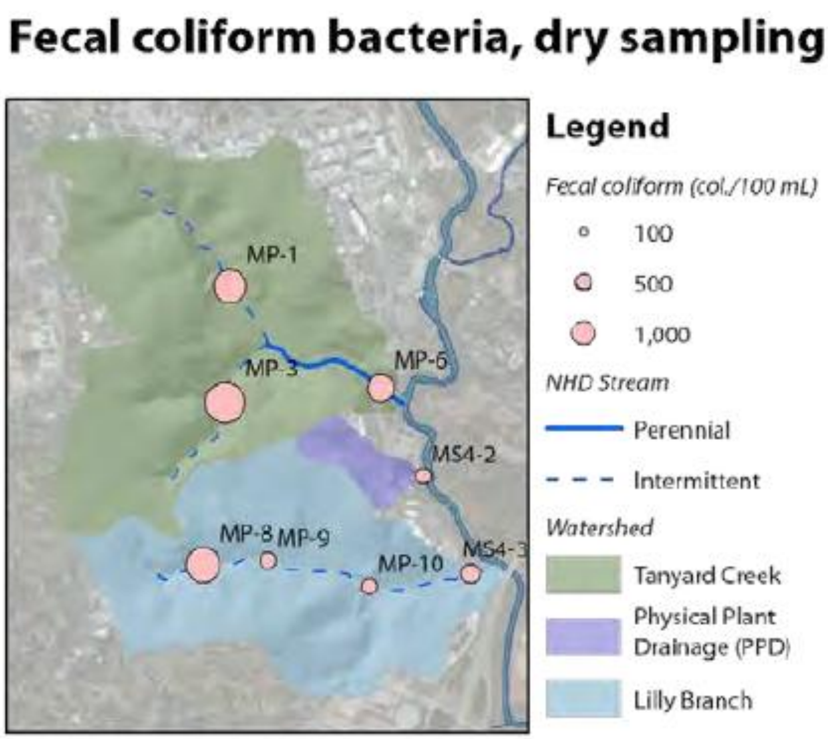


Table 8.1 Solids Accumulation and Associated Pollutant Concentrations in Urban Areas

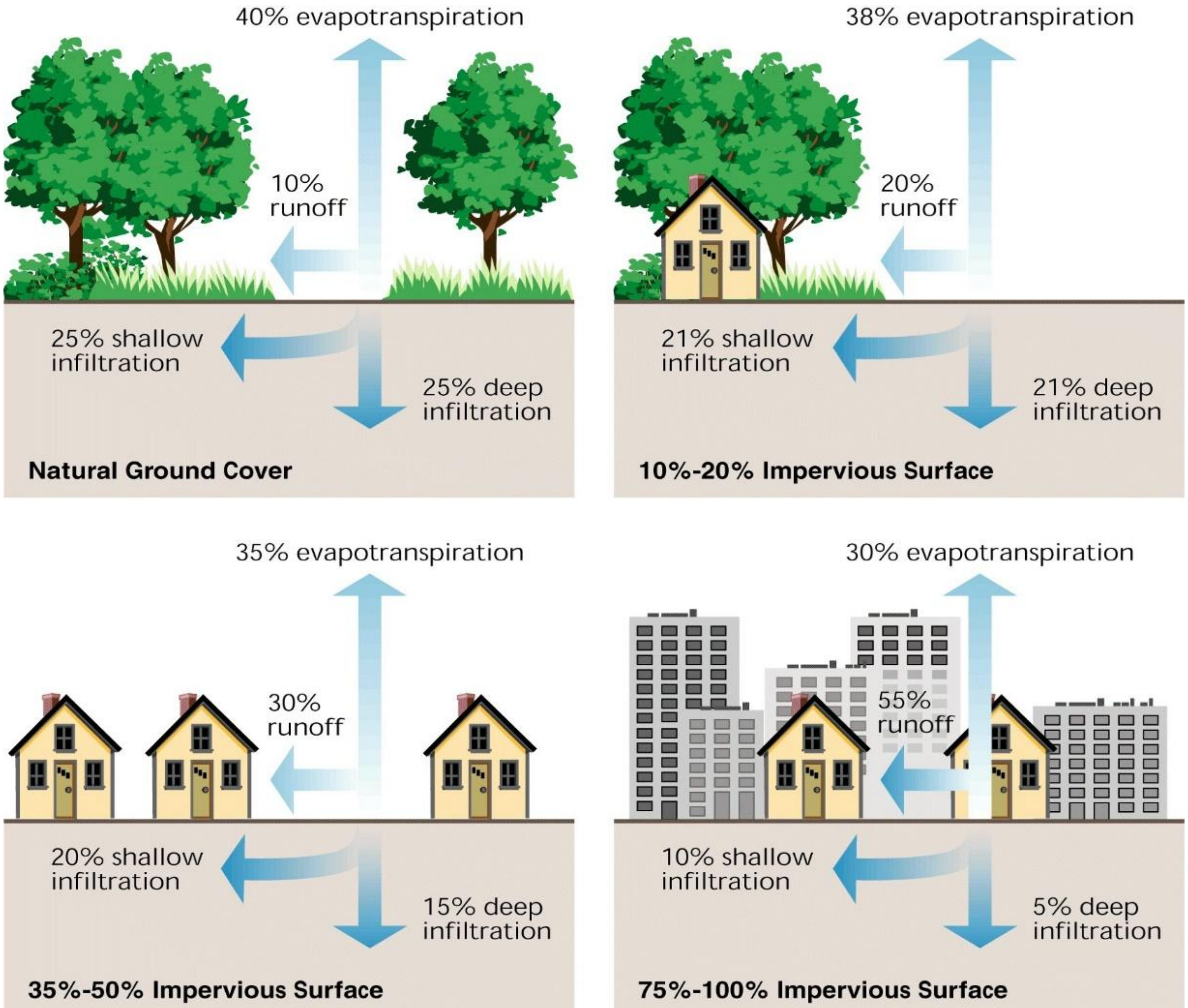
Land Use	Residential Low Density	Residential High Density	Commercial	Light Industrial	Highways
Solids accumulation (g/curb m)	10–182	30–210	13–180	80–288	13–1100
Pollutant concentration ($\mu g/g$)					
BOD_5	5,260	3,370	7,190	2,920	2,300–10,000
COD	39,300–40,000	40,000–42,000	39,000–61,730	25,100	53,650–80,000
Total N	460–480	530–610	410–420	430	223–1,600
Cd	3.2	2.7	2.9	3.6	2.1–10.2
Fecal coliforms (MPN/g)	60,570–82,500	25,621–31,800	36,900	30,700	18,768–38,000

Source: After Ellis (1986).

Table 8.8 Vehicle Emissions of Pollutants

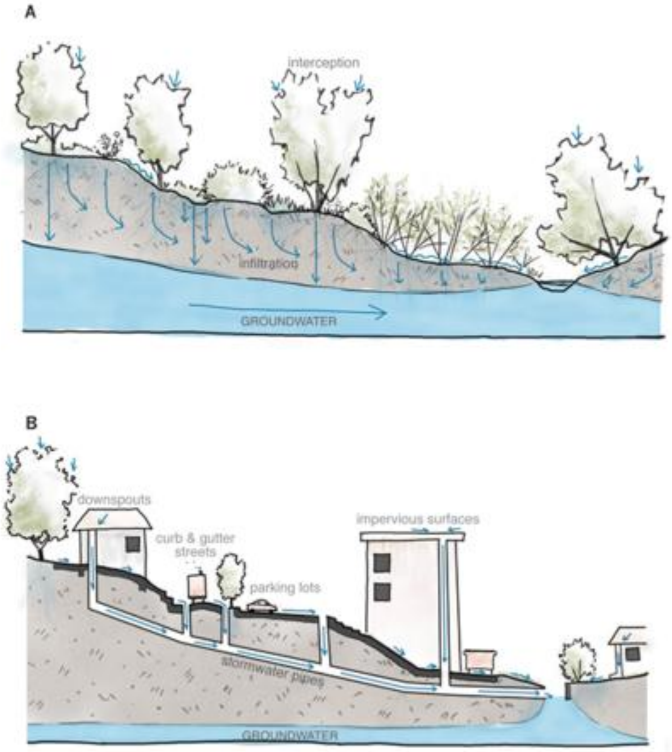
Pollutant	Emission Rate (g/km · vehicle[a]) Gasoline		Emission Rate (g/km · vehicle[a]) Diesel
Total hydrocarbons[b]	10		1
Nitric oxides[b]	3		6
Lead[b]	0.01		0
Benzo[*a*]pyrene[b]	7×10^{-7}		2×10^{-6}
Fluoranthene[b]	2×10^{-5}		4×10^{-5}
Suspended solids[c]		1.3	
Total phosphorus[c]		0.001	
Zinc[b]		0.003 (total)	
Cadmium[b]		1×10^{-6} (total)	
Copper[b]		4.5×10^{-5} (total)	
Pyrene[d]			
Without catalyst		0.003	
With catalyst		0.0015	
Phenanthrene[d]			
Without catalyst		0.002	
With catalyst		0.00015	

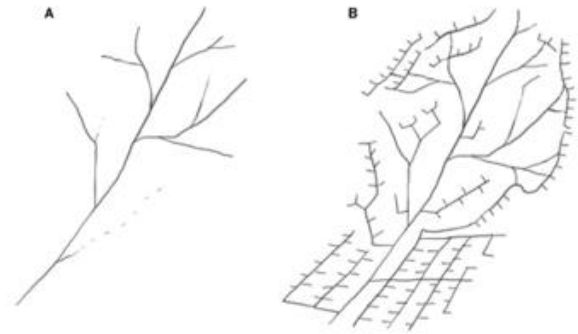
[a]Same as mg/m · vehicle.
[b]Compilation by Ball et al. (1991).
[c]Data from Shaheen (1975).
[d]Data from Acres (1991).



General stormwater principles

- The most effective runoff quality controls reduce both the volume and peak of runoff. How might this be done?
- Small storms are generally most important from a **quality** standpoint because they cause considerable washoff of pollutants without much dilution. Stormwater systems are generally designed to deal with larger storms and may not be effective for quality control.
- A rule of thumb is that SCMs and NI should limit the peak runoff of small storms less than two-years return period in addition to less frequent design storms.
- Most important pollutants in urban runoff can be settled out, but sometimes the dissolved form is also important. Control of dissolved pollutants can be more difficult than for adsorbed forms which settle out.
- Applicability of SCMs and NI is site specific.
- Longevity is a problem for some urban BMPs and NI features that tend to fill up with sediment and clog.





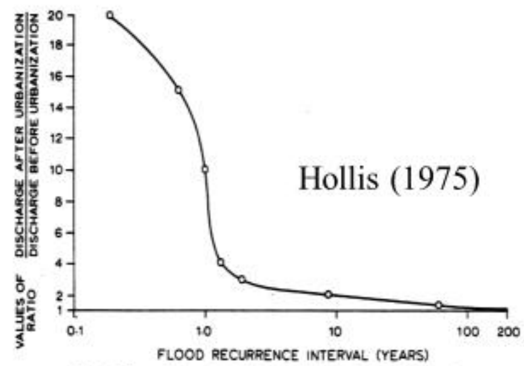
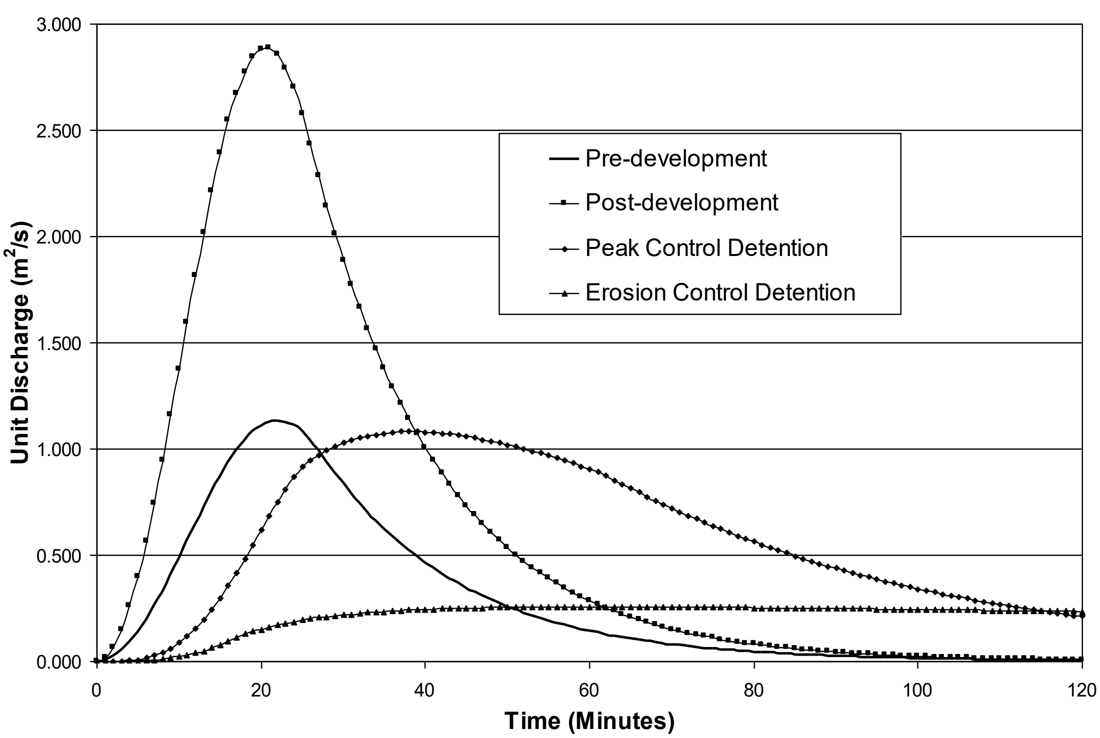
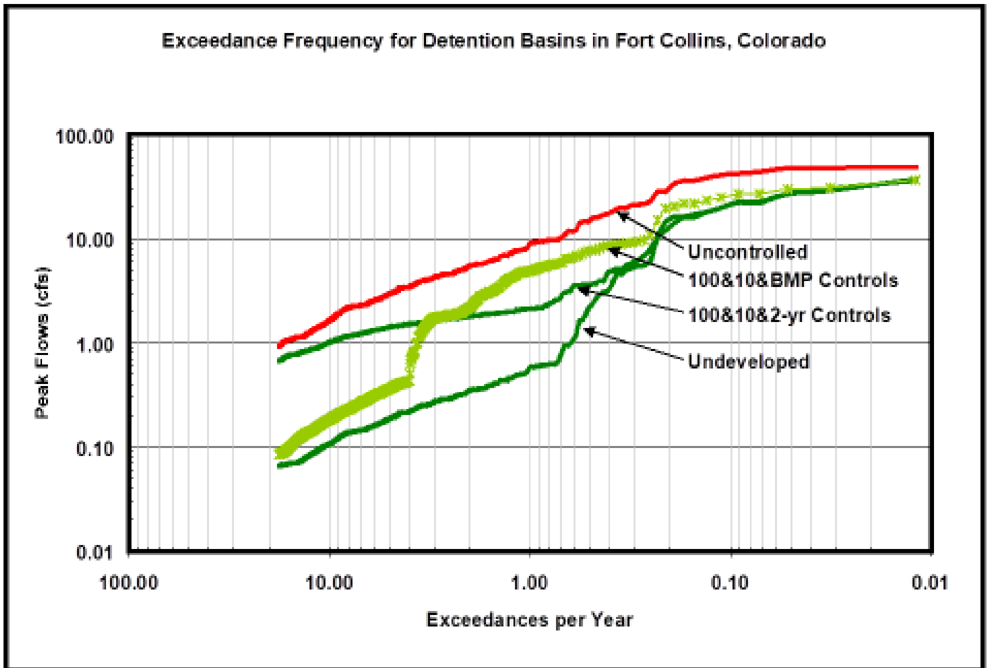


Fig. 3. Effect on flood magnitudes of paving 20% of a basis (based on Figure 2).

If you don't eat the energy, the energy will eat the channel!

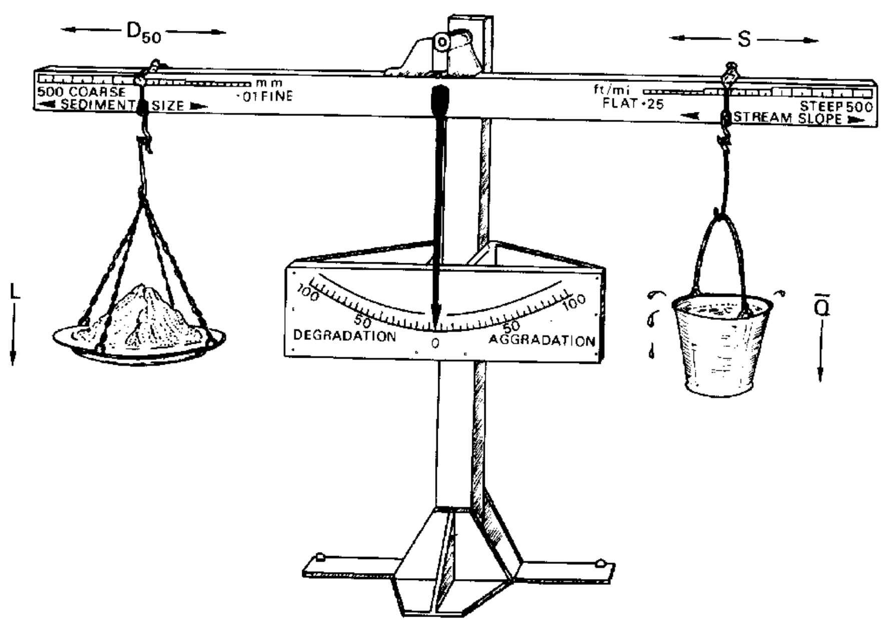
Mail Creek
Unit Discharge (2 year)





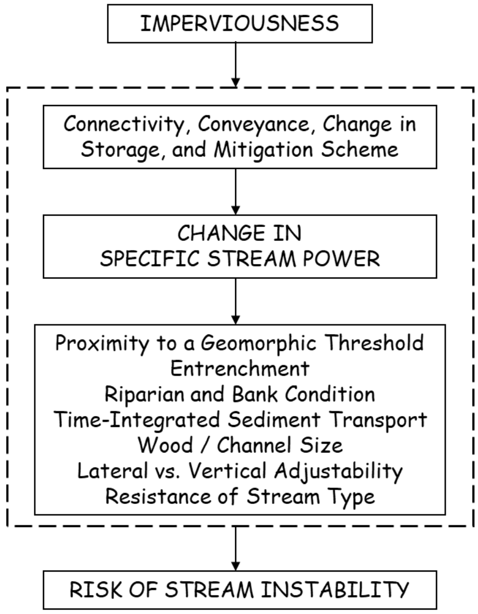
Note: BMP is "extended detention with 24-hr drawdown time.

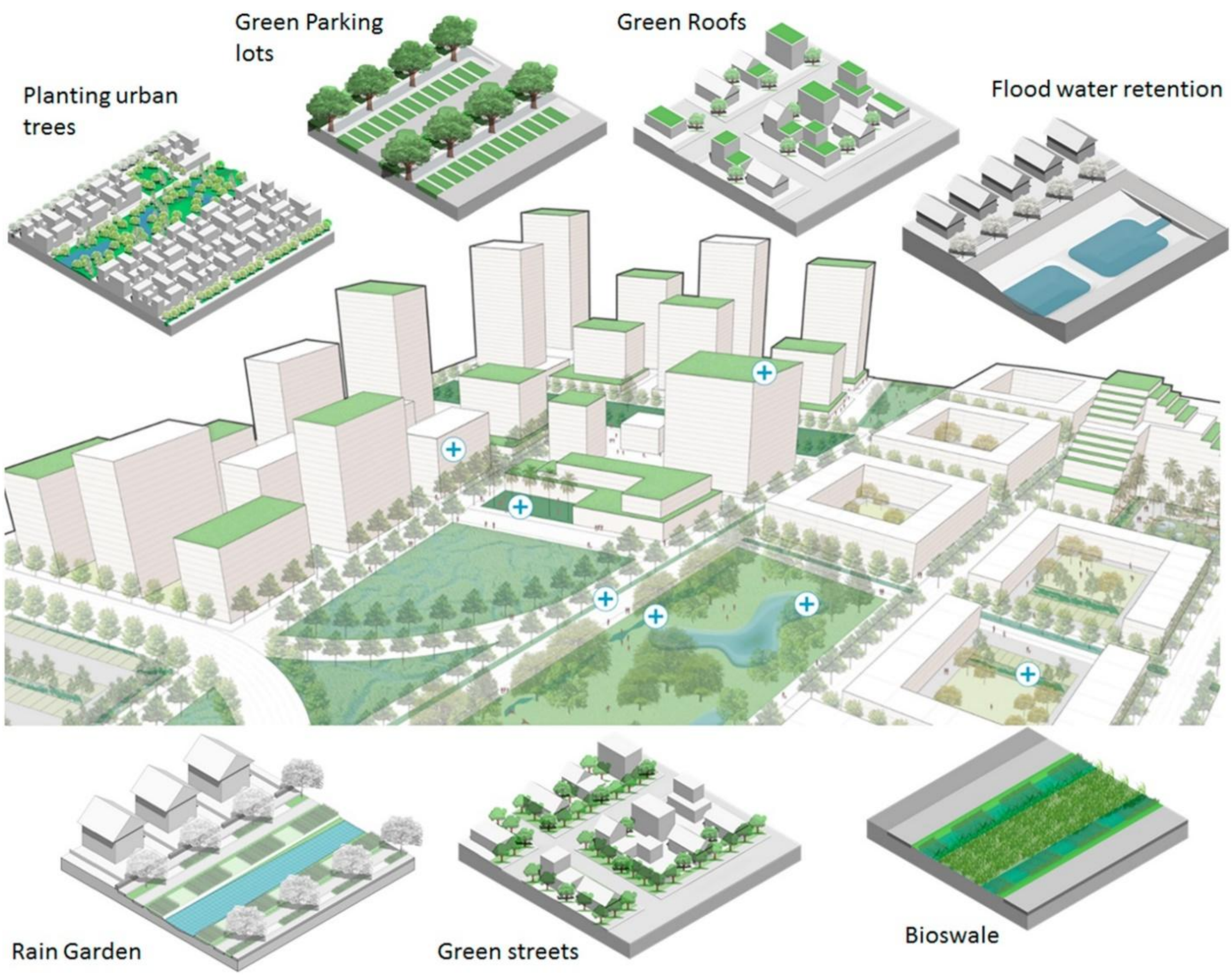
Effect of Detention and BMPs on the Post-development Flow Frequency Curve



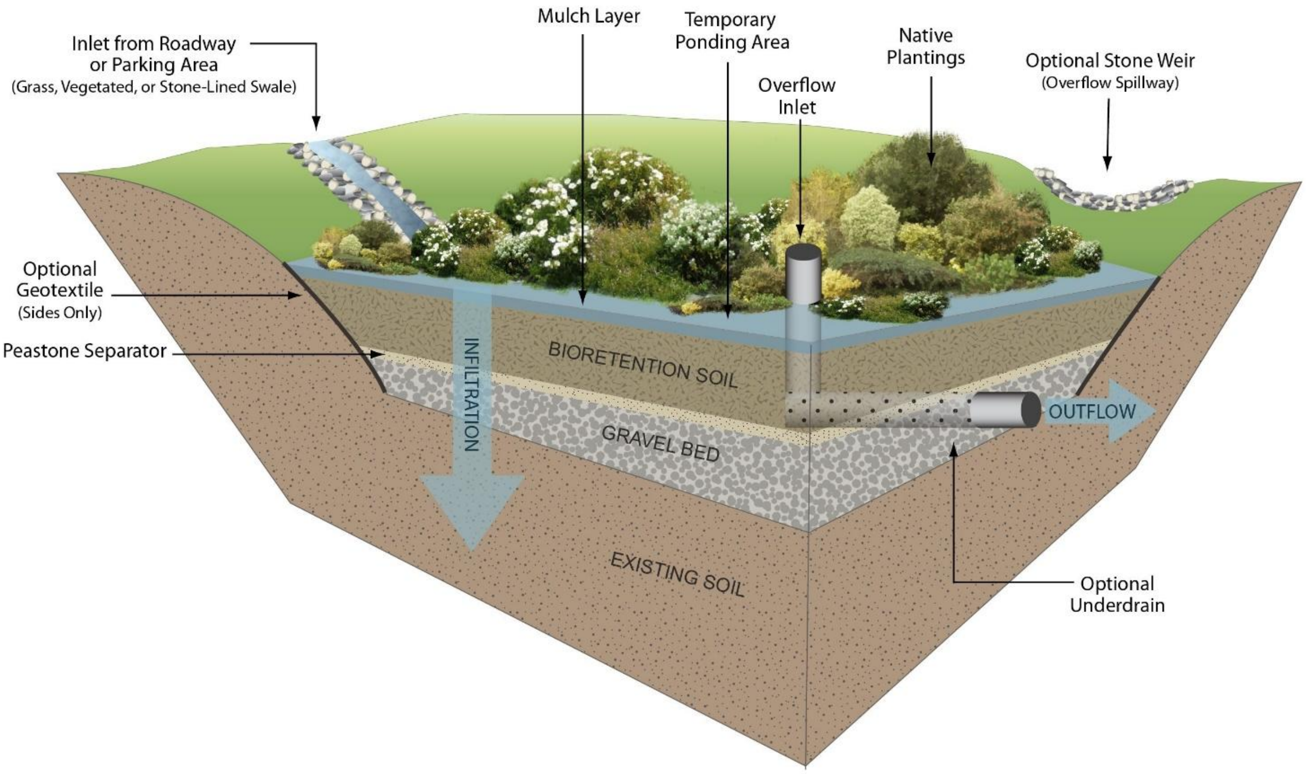
Must consider resistance of boundary materials and potential for reduced sediment load over time

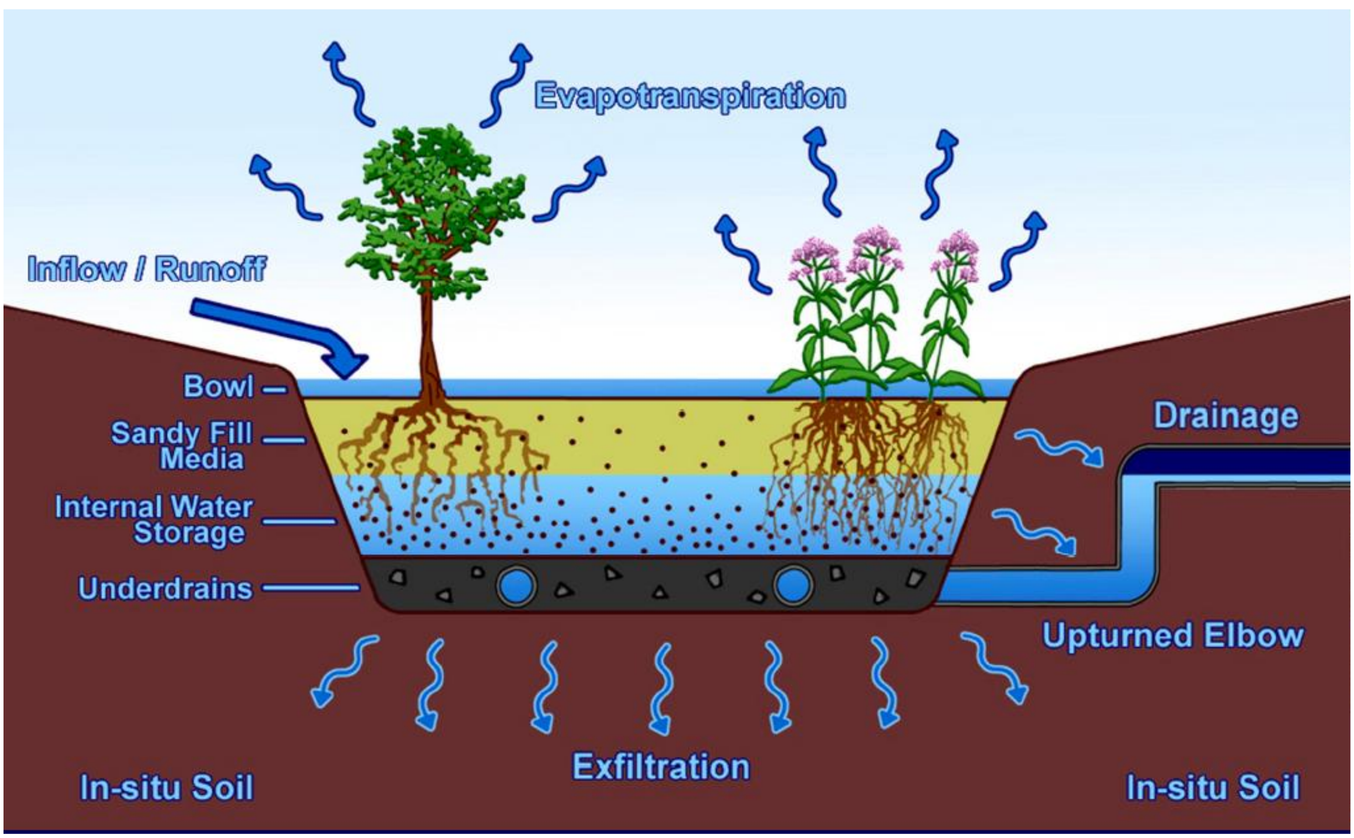
Must quantify bucket: all erosive flows over time (magnitude and duration)





Whelchel et al. (2018)


Inlet from Roadway
or Parking Area
(Grass, Vegetated, or Stone-Lined Swale)
Mulch Layer
Temporary
Ponding Area
Overflow
Inlet
Native
Plantings
Optional Stone Weir
(Overflow Spillway)
Optional
Geotextile
(Sides Only)
Peastone Separator
INFILTRATION
BIORETENTION SOIL
GRAVEL BED
OUTFLOW
EXISTING SOIL
Optional
Underdrain


Evapotranspiration
Inflow / Runoff
Bowl
Sandy Fill
Media
Internal Water
Storage
Underdrains
Drainage
Upturned Elbow
Exfiltration
In-situ Soil
In-situ Soil

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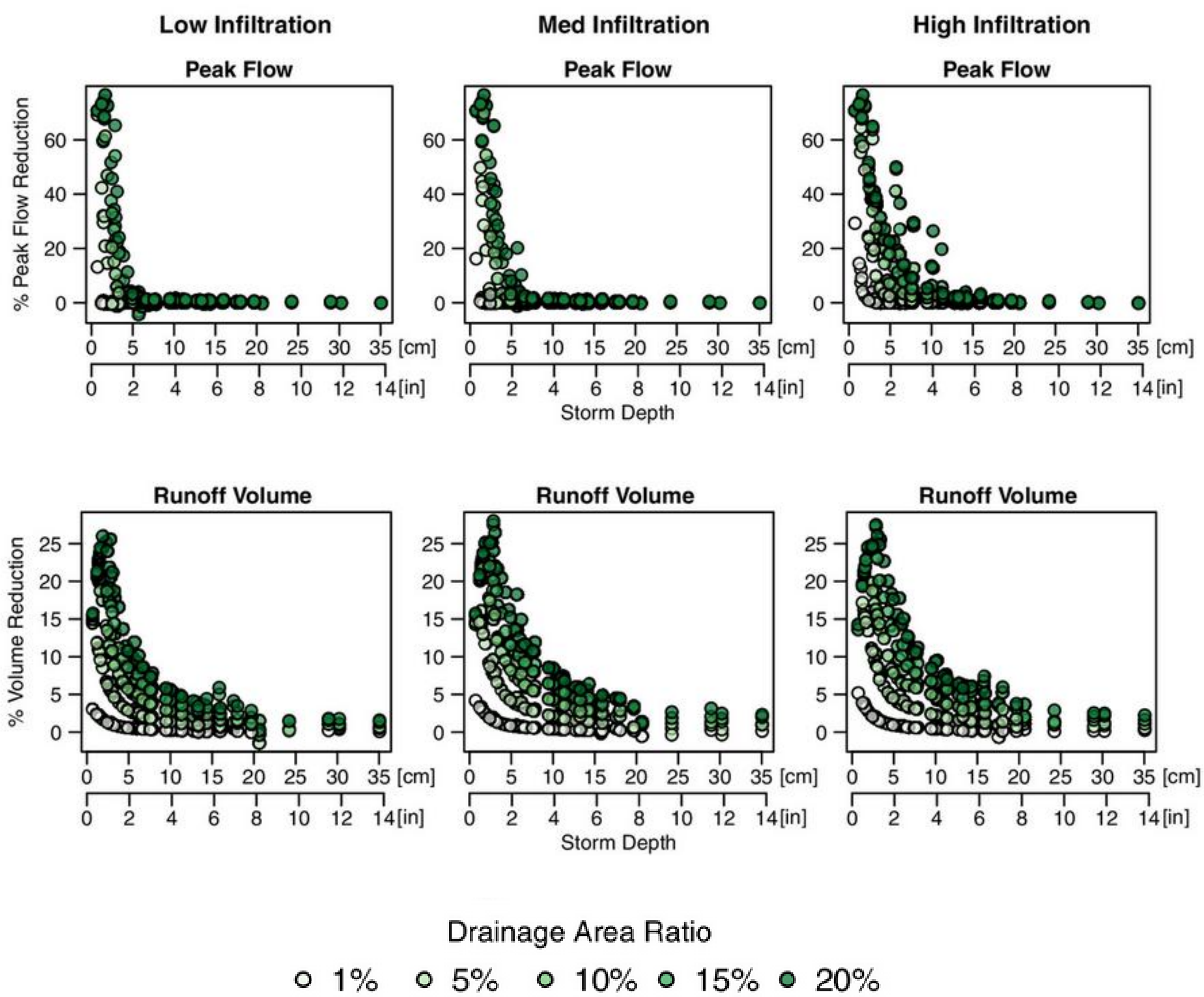
Effects of Design and Climate on Bioretention Effectiveness for Watershed-Scale Hydrologic Benefits

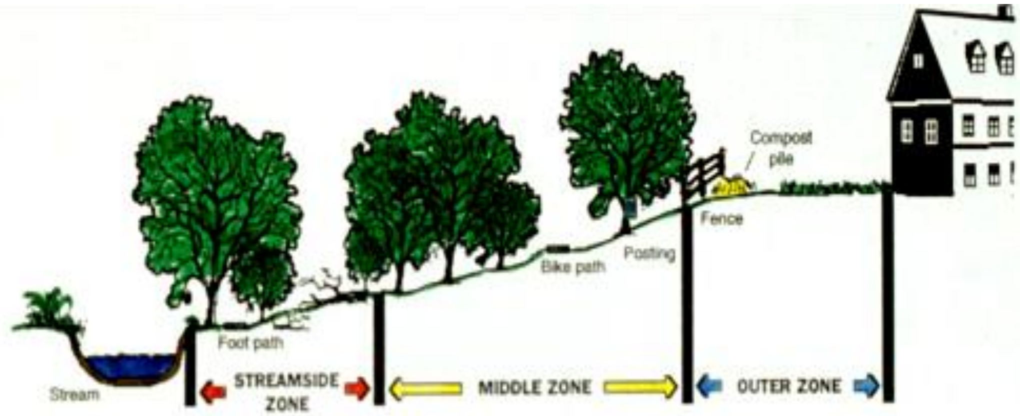
Authors: Roderick W. Lammers, Ph.D., Laura Miller, and Brian P. Bledsoe, Ph.D., M.ASCE | AUTHOR AFFILIATIONS

Publication: Journal of Sustainable Water in the Built Environment • Volume 8, Issue 4 • https://doi.org/10.1061/JSWBAY.0000993



Feature article from Watershed Protection Techniques. 1(4): 155-163

The Architecture of Urban Stream Buffers

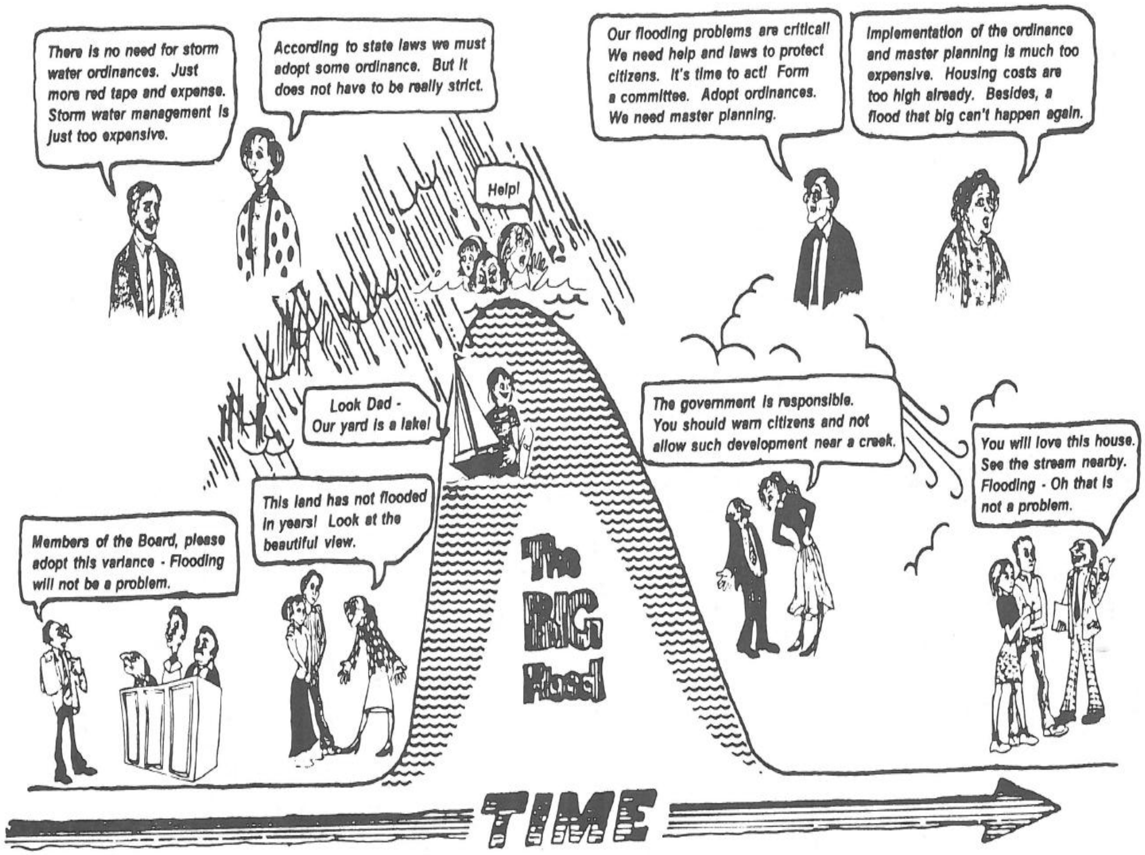


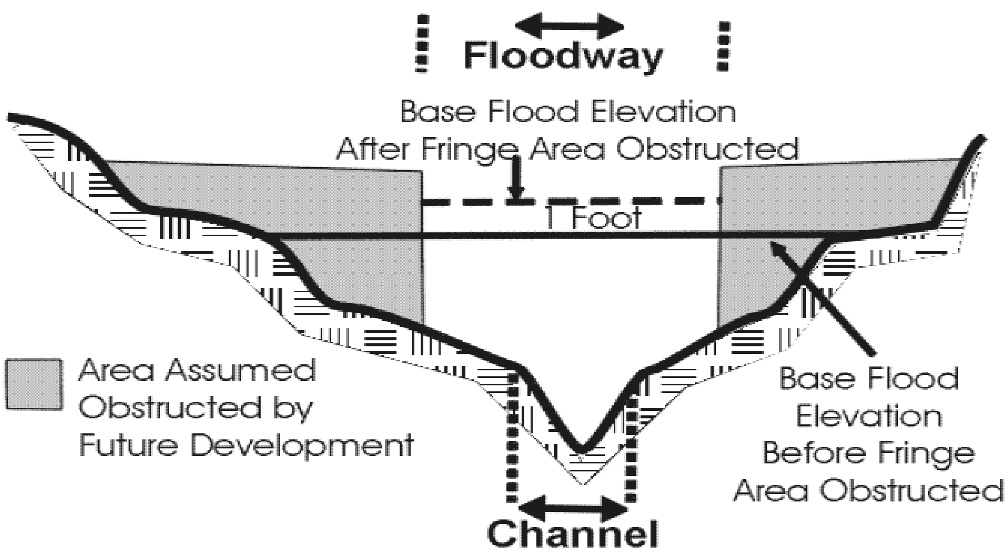
CHARACTERISTICS	STREAMSIDE ZONE	MIDDLE ZONE	OUTER ZONE
FUNCTION	Protect the physical integrity of the stream ecosystem	Provide distance between upland development and streamside zone	Prevent encroachment and filter backyard runoff
WIDTH	Min. 25 feet, plus wetlands and critical habitats	50 to 100 feet, depending on stream order, slope, and 100 year floodplain	25 foot minimum setback to structures
VEGETATIVE TARGET	Undisturbed mature forest. Reforest if grass	Managed forest, some clearing allowable	Forest encouraged, but usually turfgrass
ALLOWABLE USES	***Very Restricted*** e.g., flood control, utility right of ways, footpaths, etc.	***Restricted*** e.g., some recreational uses, some stormwater BMPs, bike paths, tree removal by permit	***Unrestricted*** e.g., residential uses including lawn, garden, compost, yard wastes, most stormwater BMPs

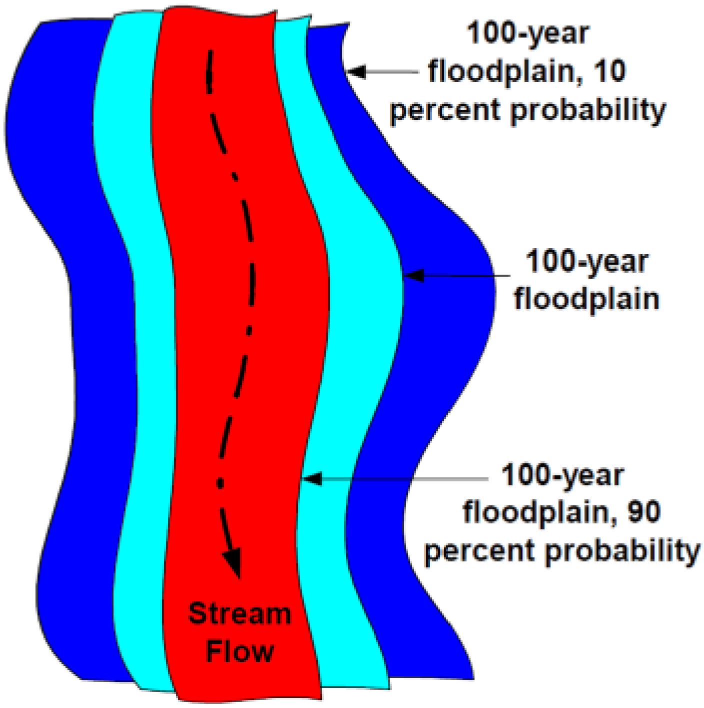
Table 2: Nuts and Bolts of an Urban Stream Buffer
■ Minimum total width of 100 feet, including floodplain
■ Zone-specific goals and restrictions for the outer, middle, and streamside zones
■ Adopt a vegetative target based on predevelopment plant community
■ Expand the width of the middle zone to pick up wetlands,slopes and larger streams
■ Use clear and measurable criteria to delineate the origin and boundaries of the buffer
■ The number and conditions for stream and buffer crossings should be limited
■ The use of buffer for stormwater runoff treatment should be carefully prescribed
■ Buffer boundaries should be visible before, during, and after construction
■ Buffer education and enforcement are needed to protect buffer integrity
■ Buffer administration should be flexible and fair to landowners

Table 1: Twenty Benefits of Urban Stream Buffers
(*f*) = Benefit Amplified by or Requires Forest Cover

1. **Reduces watershed imperviousness by 5%.** An average buffer width of 100 feet protects up to 5% of watershed area from future development.
2. **Distances areas of impervious cover from the stream.** More room is made available for placement of stormwater practices, and septic system performance is improved. (*f*)
3. **Reduces small drainage problems and complaints.** When properties are located too close to a stream, residents are likely to experience and complain about backyard flooding, standing water, and bank erosion. A buffer greatly reduces complaints.
4. **Stream "right of way" allows for lateral movement.** Most stream channels shift or widen over time; a buffer protects both the stream and nearby properties.
5. **Effective flood control.** Other, expensive flood controls not necessary if buffer includes the 100-yr floodplain.
6. **Protection from streambank erosion.** Tree roots consolidate the soils of floodplain and stream banks, reducing the potential for severe bank erosion. (*f*)
7. **Increases property values.** Homebuyers perceive buffers as attractive amenities to the community. 90% of buffer administrators feel buffers have a neutral or positive impact on property values. (*f*)
8. **Increased pollutant removal.** Buffers can provide effective pollutant removal for development located within 150 feet of the buffer boundary, when designed properly.
9. **Foundation for present or future greenways.** Linear nature of the buffer provides for connected open space, allowing pedestrians and bikes to move more efficiently through a community. (*f*)
10. **Provides food and habitat for wildlife.** Leaf litter is the base food source for many stream ecosystems; forests also provides woody debris that creates cover and habitat structure for aquatic insects and fish. (*f*)
11. **Mitigates stream warming.** Shading by the forest canopy prevents further stream warming in urban watersheds. (*f*)
12. **Protection of associated wetlands.** A wide stream buffer can include riverine and palustrine wetlands that are frequently found along the stream corridor.
13. **Prevent disturbance to steep slopes.** Removing construction activity from these sensitive areas is the best way to prevent severe rates of soil erosion. (*f*)
14. **Preserves important terrestrial habitat.** Riparian corridors are important transition zones, rich in species. A mile of stream buffer can provide 25-40 acres of habitat area. (*f*)
15. **Corridors for conservation.** Unbroken stream buffers provide "highways" for migration of plant and animal populations. (*f*)
16. **Essential habitat for amphibians.** Amphibians require both aquatic and terrestrial habitats and are dependent on riparian environments to complete their life cycle. (*f*)
17. **Fewer barriers to fish migration.** Chances for migrating fish are improved when stream crossings are prevented or carefully planned.
18. **Discourages excessive storm drain enclosures/channel hardening.** Can protect headwater streams from extensive modification.
19. **Provides space for stormwater ponds.** When properly placed, structural stormwater practices within the buffer can be an ideal location for stormwater practices that remove pollutants and control flows from urban areas.
20. **Allowance for future restoration.** Even a modest buffer provides space and access for future stream restoration, bank stabilization, or reforestation.

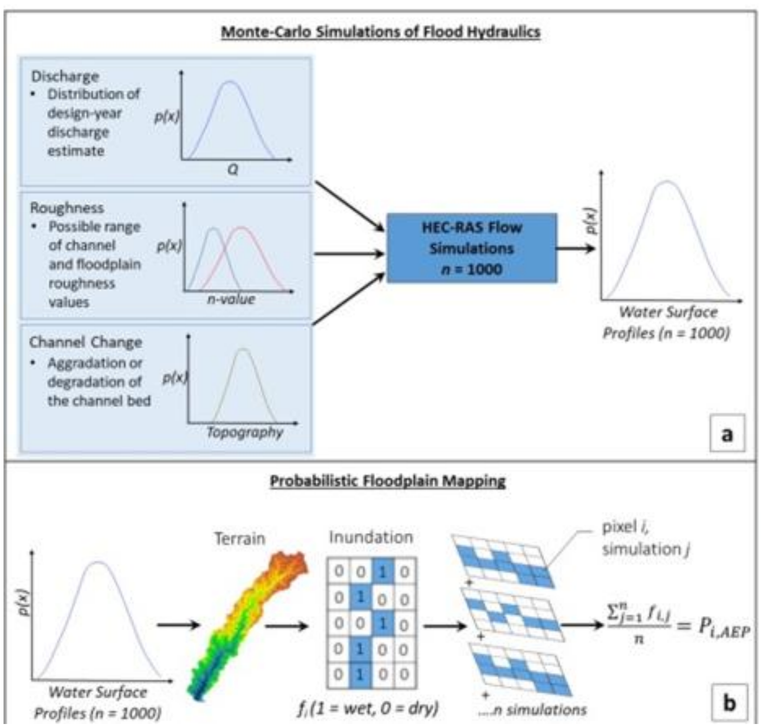

Members of the Board, please adopt this variance - Flooding will not be a problem.
There is no need for storm water ordinances. Just more red tape and expense. Storm water management is just too expensive.
This land has not flooded in years! Look at the beautiful view.
According to state laws we must adopt some ordinance. But it does not have to be really strict.
Look Dad - Our yard is a lake!
Help!
The BIG Flood
Our flooding problems are critical! We need help and laws to protect citizens. It's time to act! Form a committee. Adopt ordinances. We need master planning.
The government is responsible. You should warn citizens and not allow such development near a creek.
Implementation of the ordinance and master planning is much too expensive. Housing costs are too high already. Besides, a flood that big can't happen again.
You will love this house. See the stream nearby. Flooding - Oh that is not a problem.
TIME


Floodway
Base Flood Elevation After Fringe Area Obstructed
1 Foot
Area Assumed Obstructed by Future Development
Base Flood Elevation Before Fringe Area Obstructed
Channel

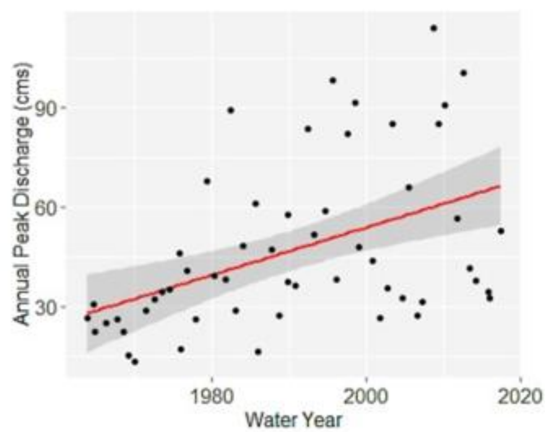


Floodplain maps that reflect uncertainty in:

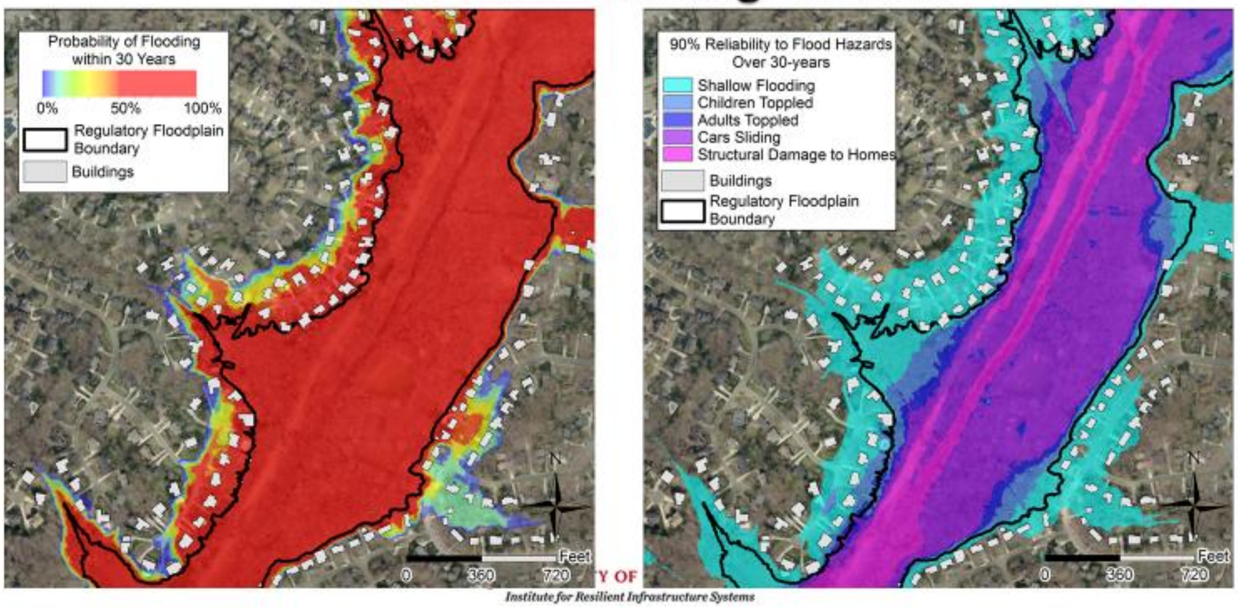
- Extreme precipitation
- Land use change
- Drainage infrastructure
- Channel geometry, channel change, sediment and debris
- Flow resistance / vegetation, debris
- Model parameters



Probabilistic Floodplain Mapping



Risk Over a Planning Horizon



Communicating flood risk:

- A 100-year flood has more than a 1 in 4 chance of occurring over a 30-year motgage.
- For 90% reliability over 30-years, structures must be above the 285-year flood.
- For 90% reliability over 50 years, structures must be above the 474-year flood.
- IF THE FUTURE BEHAVES LIKE THE PAST

Risk Ratios (Inequity > 1)

$$\textit{Race Flood Risk Ratio} = \frac{\left(\frac{At - Risk\ Black}{Total\ Black}\right)}{\left(\frac{At - Risk\ White}{Total\ White}\right)}$$

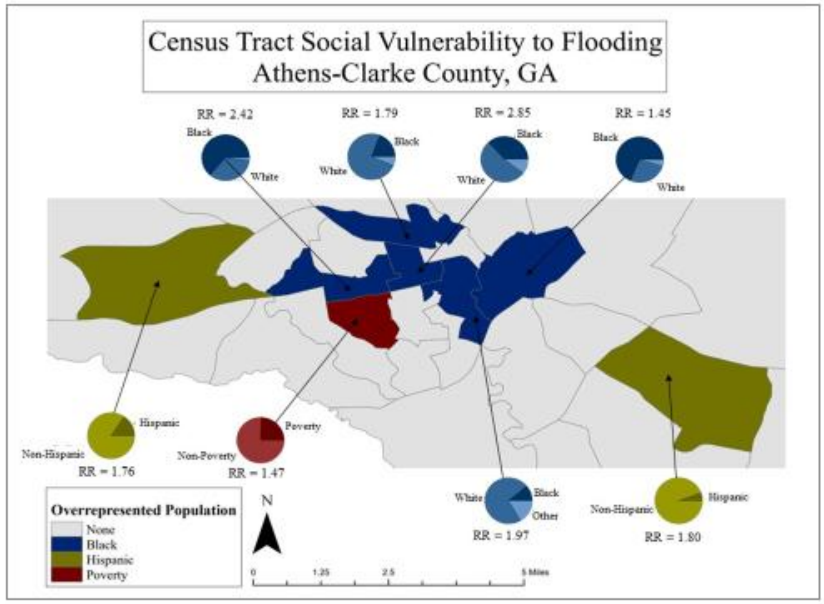
Benefit Ratios (Inequity <1)

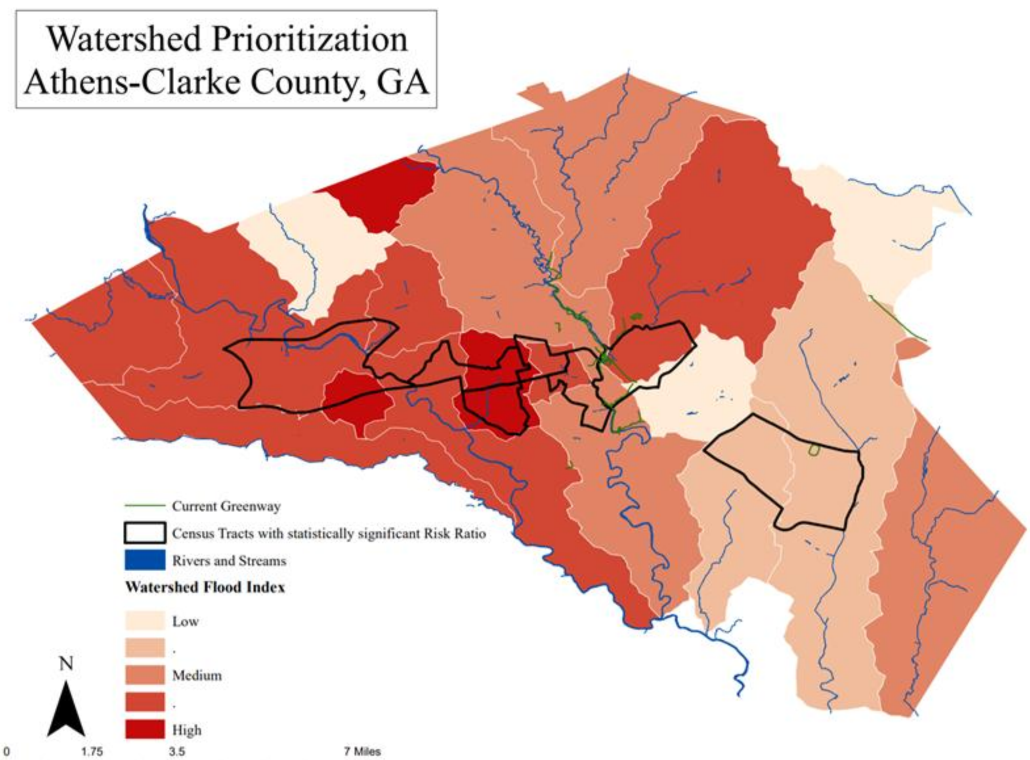
$$\textit{Poverty Benefit Ratio} = \frac{\left(\frac{Below\ Poverty\ w/Access}{Total\ Below\ Poverty}\right)}{\left(\frac{Above\ Poverty\ w/Access}{Total\ Above\ Poverty}\right)}$$

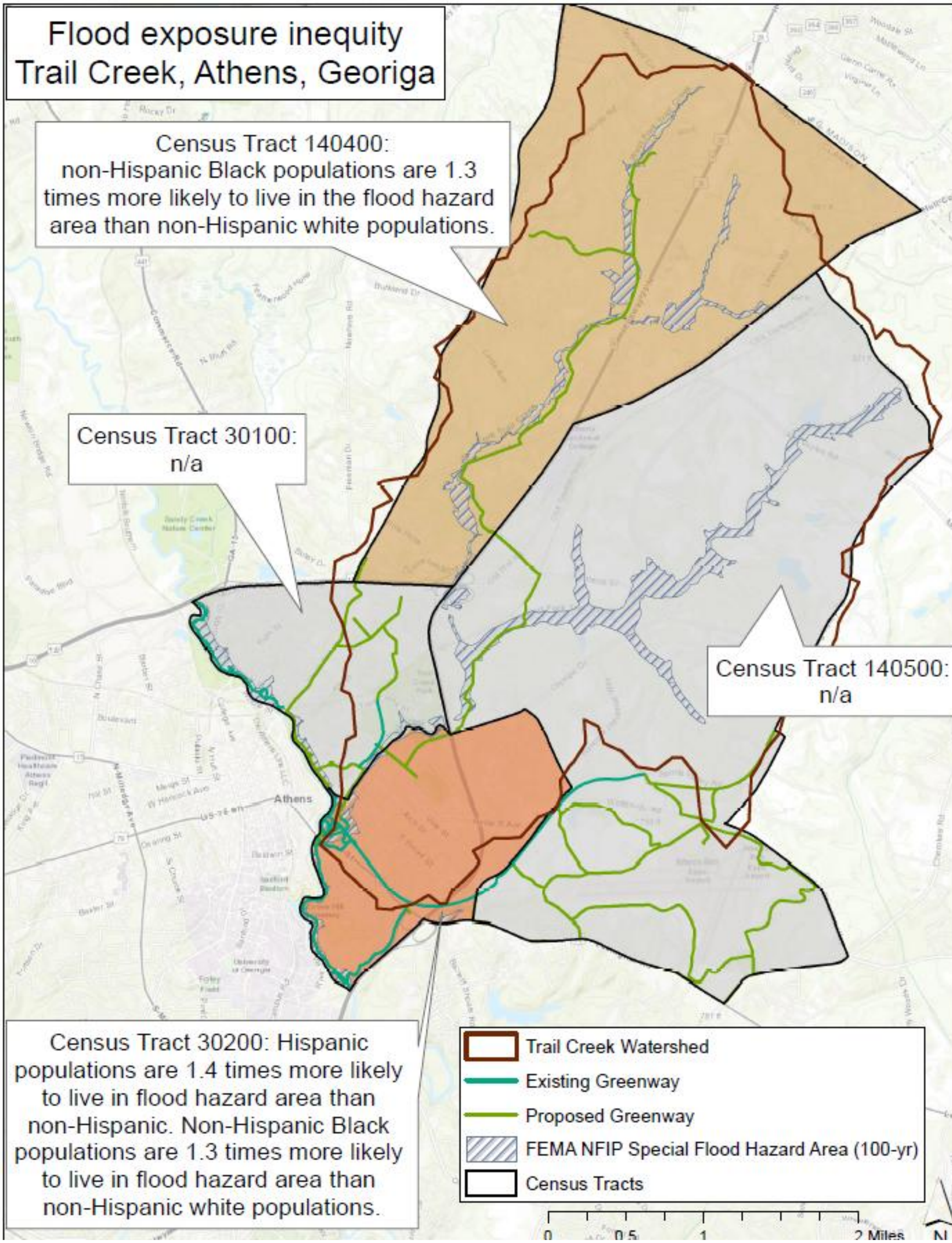
Debbage, N. A. 2019. Multiscalar spatial analysis of urban flood risk and environmental justice in the Charlanta megaregion, USA. Anthropocene 28:100226.

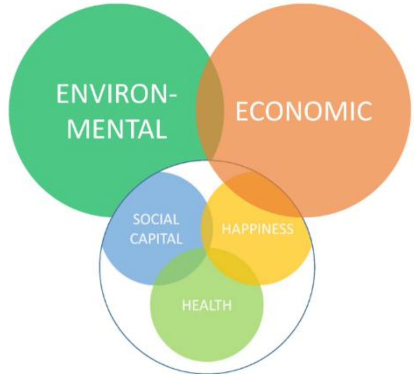
Socially vulnerable populations have higher flood exposure

Black, Hispanic, and low-income populations experience 38 to 185% greater flood risk than average in Athens, GA

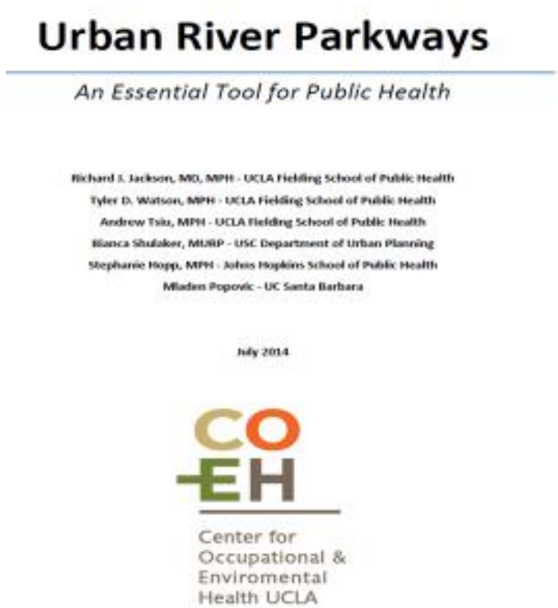





Flood exposure inequity
Trail Creek, Athens, Georiga
Census Tract 140400: non-Hispanic Black populations are 1.3 times more likely to live in the flood hazard area than non-Hispanic white populations.
Census Tract 30100: n/a
Census Tract 140500: n/a
Census Tract 30200: Hispanic populations are 1.4 times more likely to live in flood hazard area than non-Hispanic. Non-Hispanic Black populations are 1.3 times more likely to live in flood hazard area than non-Hispanic white populations.
Trail Creek Watershed
Existing Greenway
Proposed Greenway
FEMA NFIP Special Flood Hazard Area (100-yr)
Census Tracts
Athens
0
0.5
1
2 Miles
N



- Is there a role for the built environment and natural infrastructure in helping make people happier and healthier?
- This is a growing literature that suggests there is a "geography of happiness". This approach to place making aims: to make people happy, to connect them to each other, to promote good health, and to help them thrive!
- This might involve clean air, quiet, short commutes, nature contact, "third places", and beauty…
- To do this, we need to advance science and practice as well as stakeholder utility by:
    - Improving description of actual flood risks (combined effects of land use, climate, channels).
    - Identifying and addressing risk inequities.
    - Coorindating stream restoration and stormwater management to improve outcomes and define realistic expectations.
    - Scaling and targetting natural / hybrid infrastructure solutions to deliver a broader array of benefits (i.e., social, hydrological, water quality, habitat,…)
- Need to seek integration of floodplain management, stormwater (green+grey), stream and buffer restoration, urban design,…


Urban River Parkways

An Essential Tool for Public Health

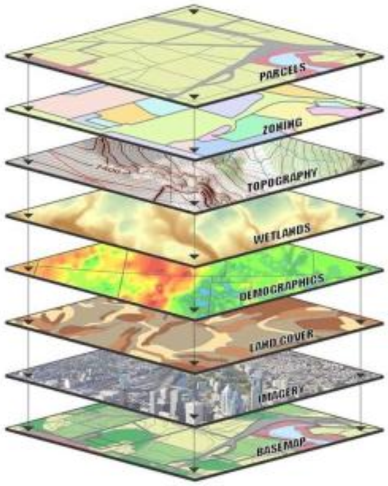
Richard J. Jackson, MD, MPH - UCLA Fielding School of Public Health
Tyler D. Watson, MPH - UCLA Fielding School of Public Health
Andrew Tsiu, MPH - UCLA Fielding School of Public Health
Bianca Shulaker, MURP - USC Department of Urban Planning
Stephanie Hopp, MPH - Johns Hopkins School of Public Health
Mladen Popovic - UC Santa Barbara

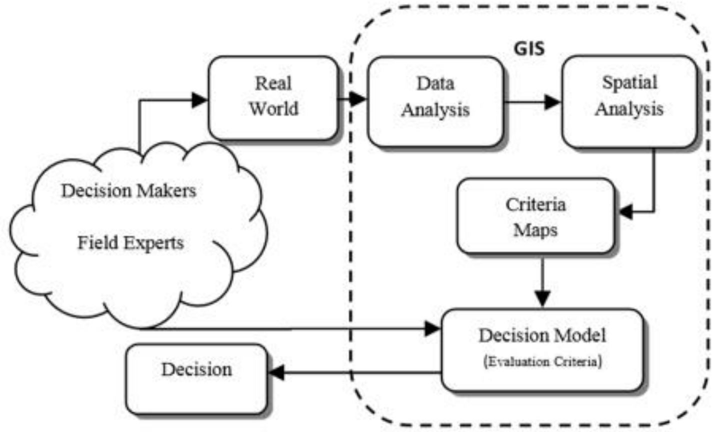
July 2014

CO
EH
Center for Occupational & Enviromental Health UCLA

Every 1 dollar spent on trails results in \$3 to >\$10 of direct medical benefit

Spatial Multi-criteria Decision Analysis (MCDA)





Ryan and Nimick (2019)

Spatial Prioritization

- Provide human and ecosystem benefits
- Choose among many challenges and potential opportunities
- Highest priority watersheds for meeting multiple objectives
- Operationalize environmental equity
- Align stream network investments
- Equitable distribution of benefits and protection from risks
- Multi-criteria decision analysis (MCDA)

Charlotte-Mecklenburg STORM WATER Services

Day 2: Exercise

Mike Wharton (ACC) and President Jere Morehead (UGA) have come to your team requesting a spatial MCDA analysis to support decision on where to allocate $10M in NI project funds in ACC and UGA, respectively. Form teams (with different people than yesterday) to identify the criteria and subcriteria that will be used to rank combinations of projects in a spatial MCDA.

Day 2: Afternoon Handout

Themes for today:

- Structured decision making is a family of methods (attempting) to make rational, transparent, and (potentially) reproducible choices. These methods INFORM decisions; people make decisions. Good decision methods don't guarantee good decision outcomes.
- Concepts of critical thinking, NBS, and structured decision-making sound great on paper, but they are really hard to execute in a constrained landscape. Ultimately, how do you "tell the story" of a project?
- Society asks water managers to simultaneously overcome past mismanagement, cope with present challenges, and adapt to future challenges, all while seeking novel solutions.

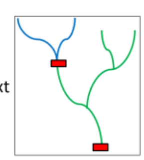
STRUCTURED DECISION MAKING:

- A structured decision making framework helps (PrOACT):
  - Define the **PR**oblem
  - Set **O**bjectives
  - Develop **A**lternatives
  - Assess **C**onsequences
  - Make **T**rade-offs
- Tips for objective setting (Gregory and Keeney's four-step process). A few prompts they also suggest that might jar some thoughts lose...List problems and opportunities. Compose a wish list. Identify a best and worst outcome. Pros/cons of good/bad alternatives. Justify the project to the public.
  - Step 1: Write down the concerns you want to address.
  - Step 2: Convert the general concerns into specific, succinct objectives. Write objectives in verb-object format.
  - Step 3: Organize objectives. Separate the ends from the means. (Ask "Why?")
  - Step 4: Clarify what is meant by each objective.
- Tips for alternative development:
  - There should be clear alignment between your alternatives and objectives.
  - Always include a "no action" alternative. This is the null hypothesis.
  - When developing alternatives, identify a range of potential actions that could be applied. Mix and match actions to create logical sets (i.e., alternatives). Consider combinability and dependency. Make sure alternatives are substantially different in terms of philosophy, cost, and benefit. Iterate on your alternatives.

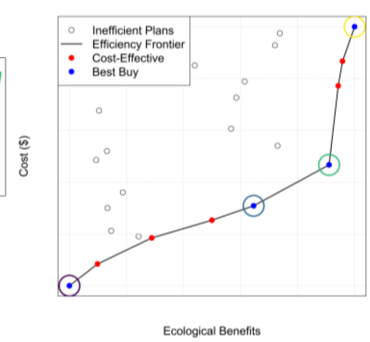
Assessing Consequences:
Summarizing Outcomes

What information does the decision maker need?

- Knowledge of the system, objectives, and decision context
- Analysis of costs, benefits, and return-on-investment
- Constraints and secondary objectives



Alternative	Cost	Ecological Benefit	Potential for Contamination?	Historical Preservation?	Failure Risk?
0: No action	0	0	No	No	High
1: Fish ladder	$$	+	No	Yes	High
2: Dam removal	$	++	Yes	No	Low
3: Removal with sediment mngmt	$$$	+++	No	No	Low



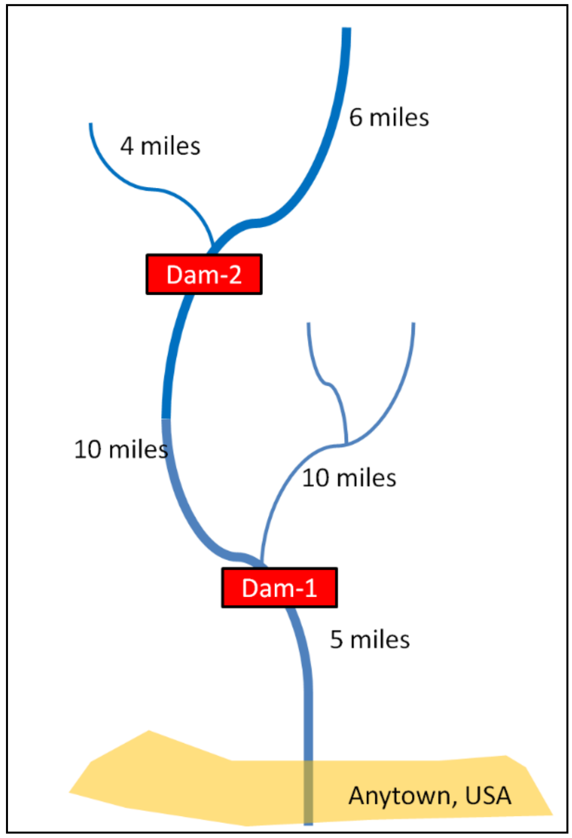
River Styx

Background:

- Small watershed on the Pacific coast
- Two obsolete mining dams, which present a flood risk to Anytown
- Commercially valuable salmon run blocked by dams
- Problem: State Environmental Agency wants to take action to reduce flood risks and benefit the environment, but funding is limited.

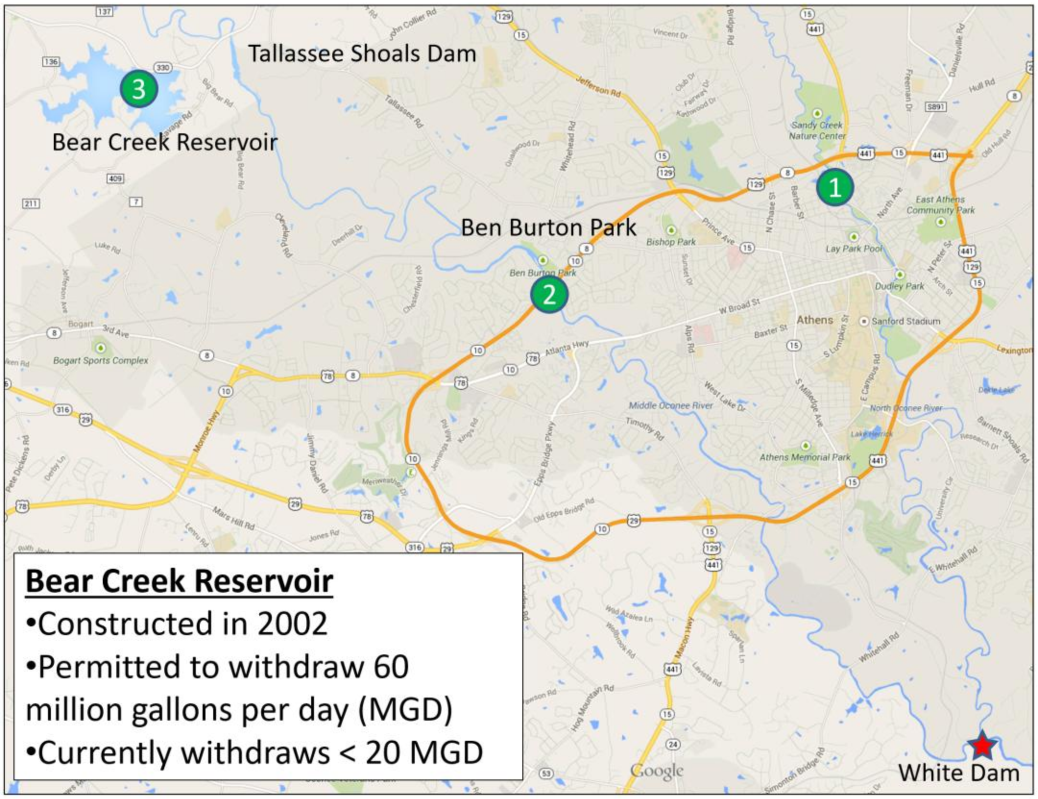
A few tasks:

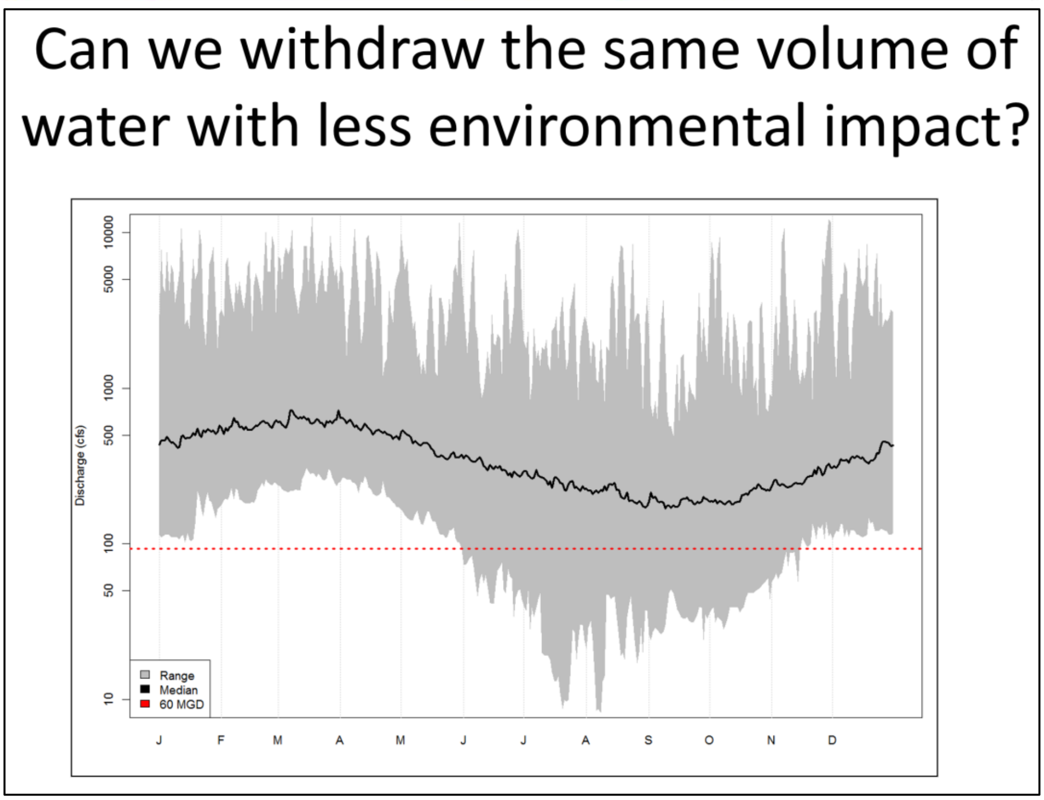
- Set at least one **objective** beyond environmental benefit.
- Consider whether there are other **alternatives** to add to the matrix.
- Identify a mechanism to assess **consequences** for the objectives.
- Populate the decision matrix to examine **trade-offs.**
- Recommend an action.



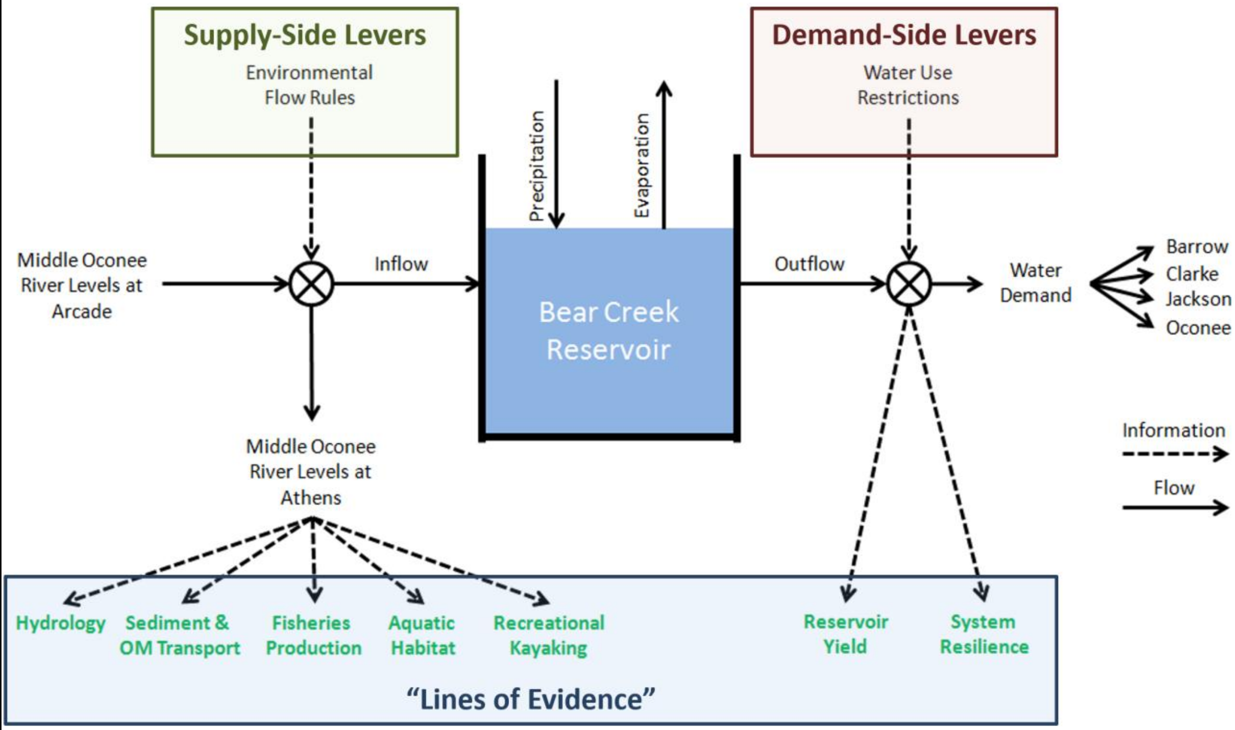
Alternative	Cost	Ecological Benefit	
No action	0		
Remove Dam-1	2,000,000		
Remove Dam-2	1,500,000		
Remove both dams	3,000,000		

Day 3: Handout

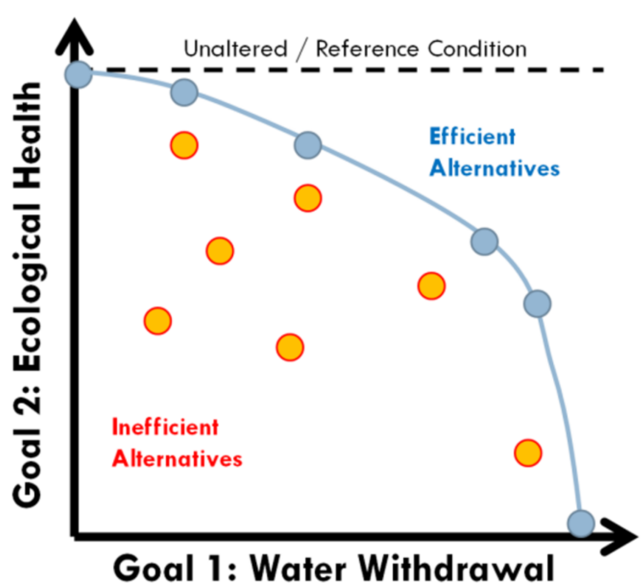


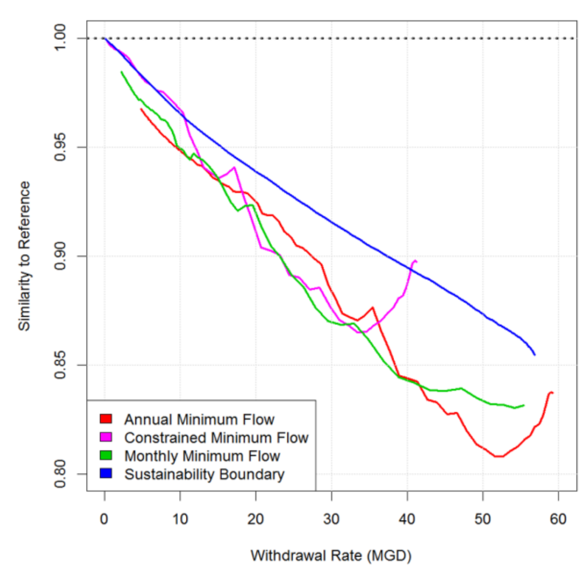


Can we meet municipal water demand with less environmental impact?



Making Trade-offs





Day 3: Hypothetical Class Project

Background (totally made up)

Athens Clarke County would like to invest in the Hunnicutt Creek watershed. The town anticipates significant growth in the area due to an upcoming change in zoning. You represent a local engineering and environmental consulting firm seeking a contract with the city to execute all site planning and stormwater management activities in the watershed. The city's request for proposals mentions the need to align with a range of community values, reduce exposure to natural hazards, and support thriving local ecosystems. The principals at your firm have given you one day and a team of 3-4 team members (an investment of ~$5,000) to scope this watershed plan. The firm wants you to outline the basic components of the broader project so that a more detailed proposal can be written. By tomorrow morning, you need to develop a presentation that applies the PrOACT framework to "tell the story" of the Hunnicutt Creek watershed. Some recommended tasks are below. You may not get to everything, and that's okay.

Problem Identification

- Explore the materials given as well as the system itself (hop in a van). Seek to build context relative to the SETS framing (social-ecological-technical systems).
    - Social context: What do you view as challenges this community is facing? Are any of those challenges caused or exacerbated by water issues?
    - Ecological context: What ecosystems are there? What are key hydrological processes? What organisms would you anticipate living here? Consider recording notes as a conceptual model.
    - Built context: Where is existing infrastructure? Is it performing as intended? What issues or hazards could be addressed by new features? How would new features interact?
- Collect sufficient notes to articulate a few problems or opportunities in the system. How are they different/similar to comparable systems elsewhere?
- As discussed, I don't believe in cookbooks, but the example statement may provide some key points to consider.

<u>Problem</u> statements: A "cookbook"?

A wants to do **X** to achieve **Y** over time **Z** in place **W** considering **B** and uncertainty in **C**.

- **A** = decision maker in charge of the decision
- **X** = action statement that indicates the scope of resources to be allocated
- **Y** = ultimate goal(s) to be achieved by doing "X"
- **Z** = temporal extent of the decision (planning horizon)
- **W** = spatial extent of the decision (refuge, management unit, etc.)
- **B** = potential constraints (legal, financial, and political)
- **C** = critically important uncertainties

Objective Setting

- Identify **at least three objectives** for watershed management.
- I find it helpful to work through Gregory and Keeney's four-step process.
    - Step 1: Write down the concerns you want to address.
    - Step 2: Convert the general concerns into specific, succinct objectives. Write objectives in verb-object format.
    - Step 3: Organize objectives. Separate the ends from the means. (Ask "Why?")
    - Step 4: Clarify what is meant by each objective.

- A few prompts they also suggest that might jar some thoughts lose...List problems and opportunities. Compose a wish list. Identify a best and worst outcome. Pros/cons of good/bad alternatives. Justify the project to the public.

Alternative Development

- Develop **at least four alternatives** for addressing water management challenges. Present a range of actions from conventional infrastructure to nature-based solutions.
- There should be clear alignment between your alternatives and objectives.
- Include a "no action" alternative. This is one of the four alternatives.
- When developing other alternatives, identify potential actions that could be applied (in a general sense). Mix and match actions to create logical sets (i.e., alternatives). Consider combinability and dependency. Make sure alternatives are substantially different in terms of philosophy, cost, and benefit. Iterate on your alternatives.

Consequences, Trade-offs, and Recommendation

- Develop a decision-matrix and an approach to assess each alternative relative to each objective. Come up with qualitative cost estimates to help compare actions.
- Think through the trade-offs associated with your alternative.
- Is there an alternative that rises to the top that you want to call your boss's attention to?

Notes for the Future

- Record any key pieces of information you felt limited by. Were there key uncertainties hindering decision making? Are there analyses that could fill gaps?

Outline a Monitoring Plan

If selected by the city and the project was built, how would you know it was successful? How would you track efficacy of outcomes? What qualitative and quantitative variables will you monitor? What is the experimental design (frequency, locations, etc.)? Who will conduct monitoring? How will data be archived and used?

Reporting

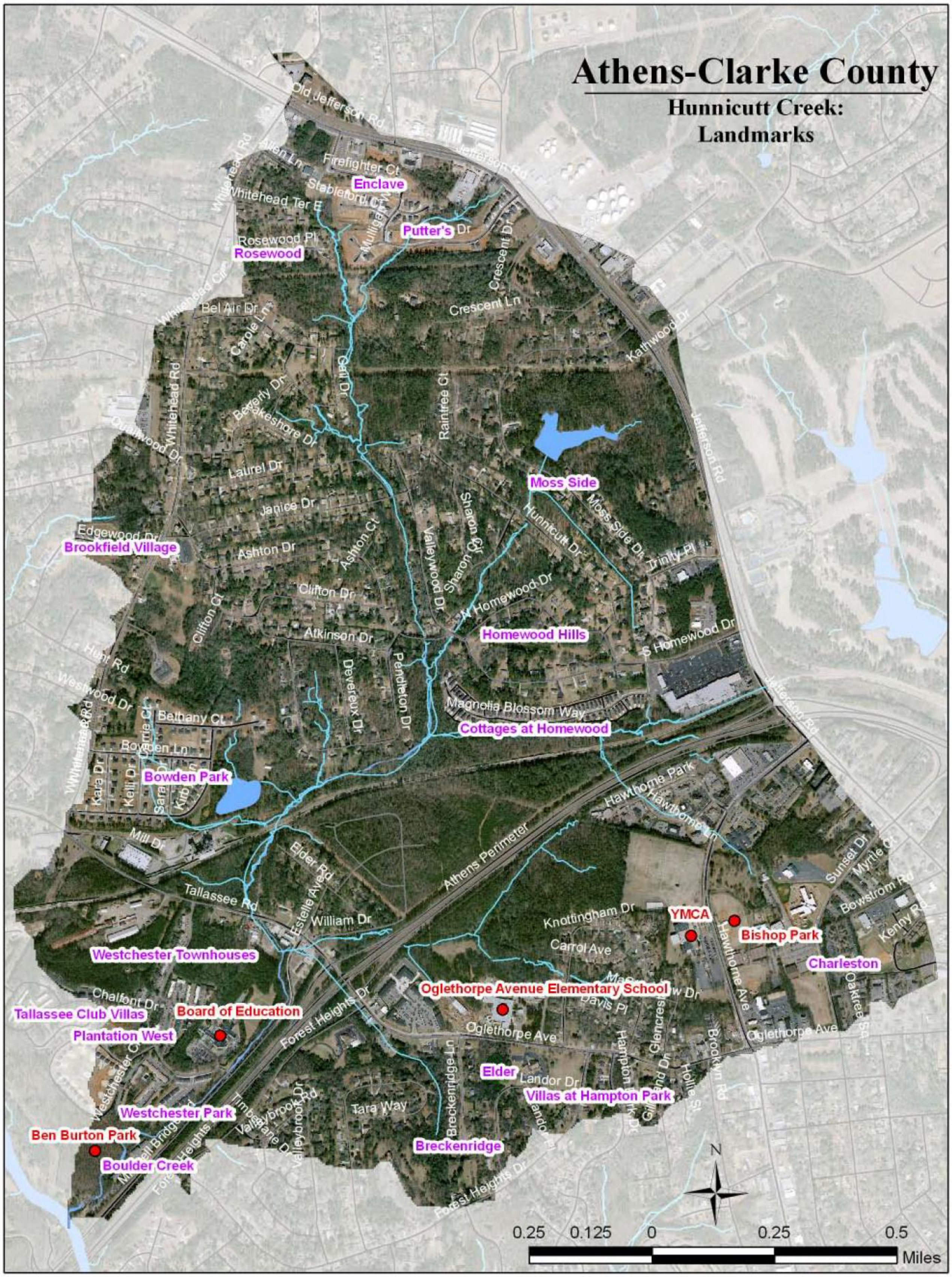
Compile information into a presentation format of your choosing. This could include slides, a poster, a narrative description, or anything else you deem appropriate.

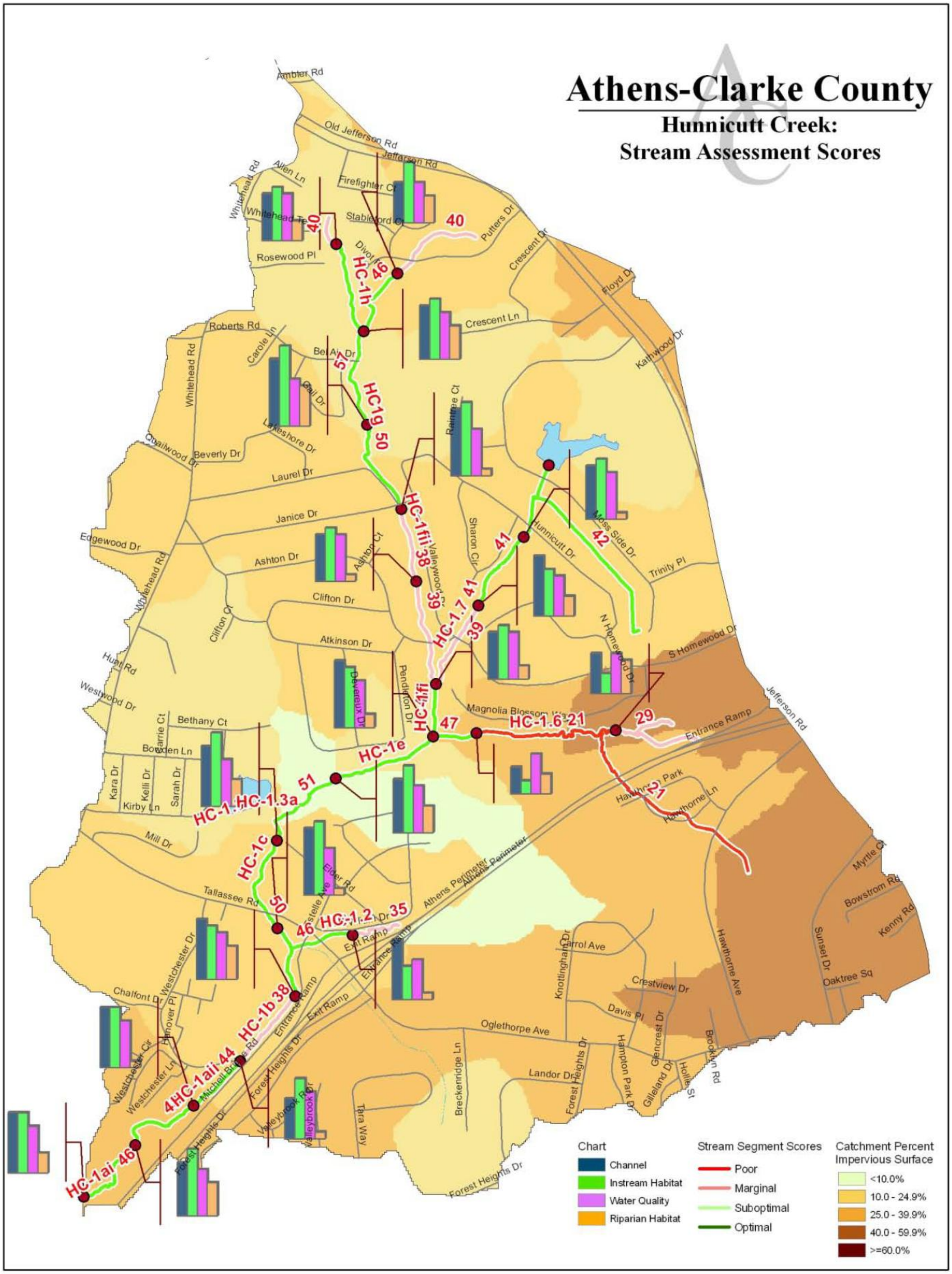
Day 3: Exercise

Hunnicutt Creek Conceptual Model

In teams of 3-4 people, develop a conceptual model of the watershed. The model can take any format (i.e., sketch, box-and-arrow, diagram), but it should seek to convey your understanding of the system and "tell the story" of the project. It may be constructive to use the steps for conceptual model development from Fischenich (2008). Discuss with your team how management objectives are featured in the model.

Steps for Conceptual Model Development (from Fischenich 2008)	**How you addressed them relative to White Dam**
1) State the model objectives.	
2) Bound the system of interest.	
3) Identify critical model components within the system of interest.	
4) Articulate the relationships among the components of interest.	
5) Represent the conceptual model.	
6) Describe an expected pattern of model behavior.	
7) Test, review, and revise as needed.	


Athens-Clarke County
Hunnicutt Creek:
Landmarks
Enclave
Putter's
Rosewood
Moss Side
Brookfield Village
Homewood Hills
Cottages at Homewood
Bowden Park
YMCA
Bishop Park
Charleston
Westchester Townhouses
Oglethorpe Avenue Elementary School
Tallassee Club Villas
Board of Education
Plantation West
Elder
Villas at Hampton Park
Westchester Park
Ben Burton Park
Boulder Creek
Breckenridge
0.25 0.125 0 0.25 0.5
Miles
N


Athens-Clarke County
Hunnicutt Creek:
Stream Assessment Scores
Chart
Channel
Instream Habitat
Water Quality
Riparian Habitat
Stream Segment Scores
Poor
Marginal
Suboptimal
Optimal
Catchment Percent Impervious Surface
<10.0%
10.0 - 24.9%
25.0 - 39.9%
40.0 - 59.9%
>=60.0%
HC-1h
HC1g 50
HC-1fii 38
HC-1.7 41
HC-1.6 21
HC-1e
HC-1.3a
HC-1c
HC-1.2
HC-1b 38
HC-1aii 44
HC-1ai 46
Oglethorpe Ave
Athens Perimeter
Jefferson Rd
Hawthorne Ave
Whitehead Rd
Tallassee Rd
Hunnicutt Dr
Crescent Ln

Day 4: Handout

- In Practice, NI is (loosely) defined by several common characteristics, not all of which may be met by any particular feature: (1) Performs infrastructure services, (2) Consists, at least in part, of natural or living materials, (3) Provides environmental and social benefits beyond typical purposes, and (4) Enhances resilience through self-adjustment.
- These characteristics can be met across infrastructure life spans:
  - Planning and design of new infrastructure
  - Operations and maintenance of existing infrastructure
  - Sunsetting and disposal of legacy infrastructure
- NI projects are complex and must fit in the context of the landscape. All NI projects have a broader social context (equity, community use, stakeholder support) and ecological context (e.g., source-sink dynamics).
- A conventional view of infrastructure pits infrastructure vs. environment, but environmental and societal outcomes are not necessarily at odds (and they are often synergistic).
- Some common themes we heard about natural infrastructure this week.
  - NI is not new and guidance exists.
  - NI is an important tool within standard engineering practice.
  - NI is not a salve for all situations, design settings, or objectives.
  - Identifying the right team is the best to build successful projects.
- Structured decision making is a family of methods (attempting) to make rational, transparent, and (potentially) reproducible choices. These methods INFORM decisions; people make decisions. Good decision methods don't guarantee good decision outcomes.
- Concepts of critical thinking, NBS, and structured decision making sound great on paper, but they are really hard to execute in a constrained landscape. Ultimately, how do you "tell the story" of a project?
- Society asks water managers to simultaneously overcome past mismanagement, cope with present challenges, and adapt to future challenges, all while seeking novel solutions.